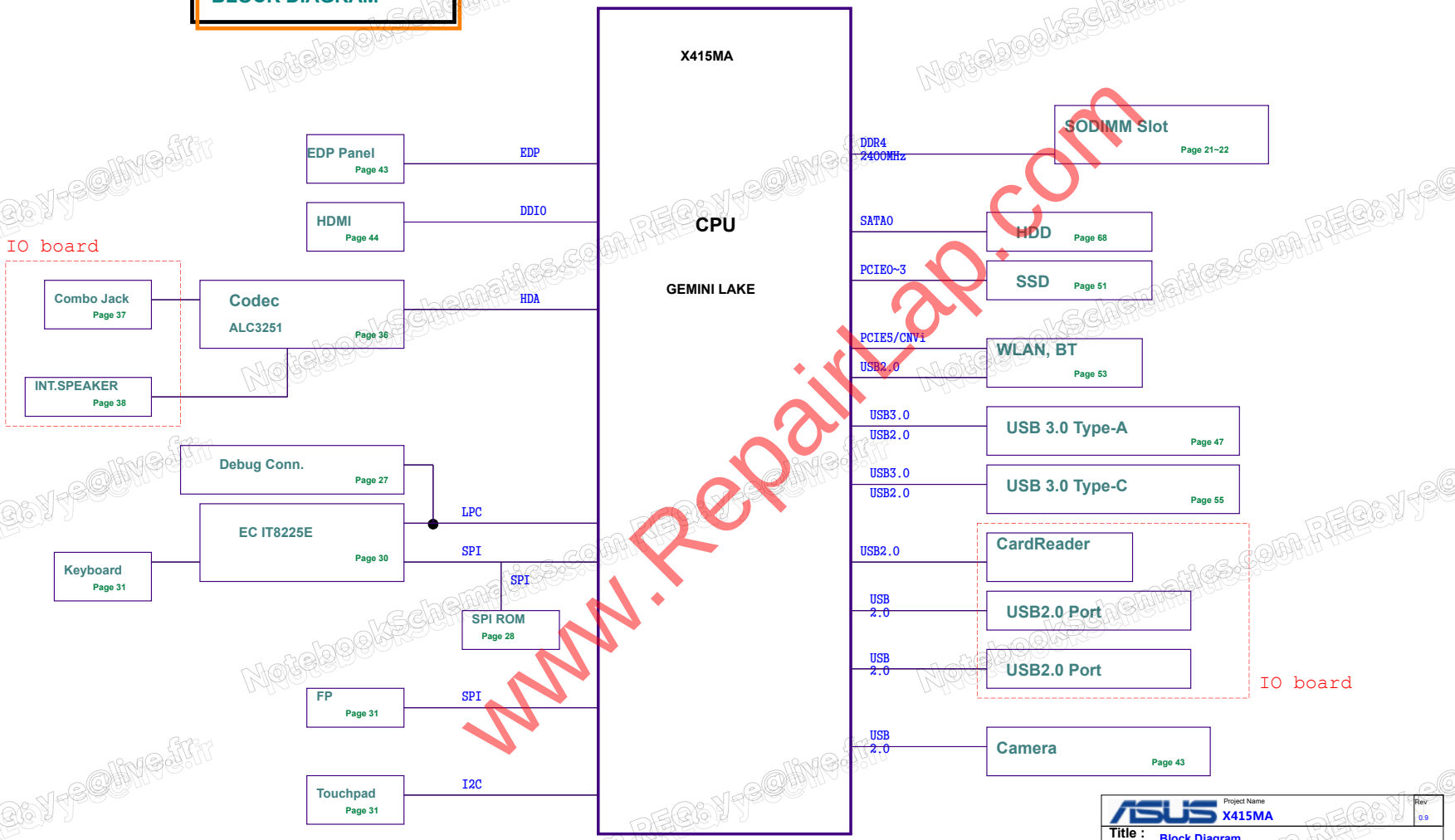
BLOCK DIAGRAM



ASUS	Project Name: X415MA		Rev: 0.9
Title : Block Diagram			
Size: B	Dept.: ASUSTeK COMPUTER INC.	Engineer: Mike	
Date: Thursday, September 10, 2020	Sheet 1	of 99	

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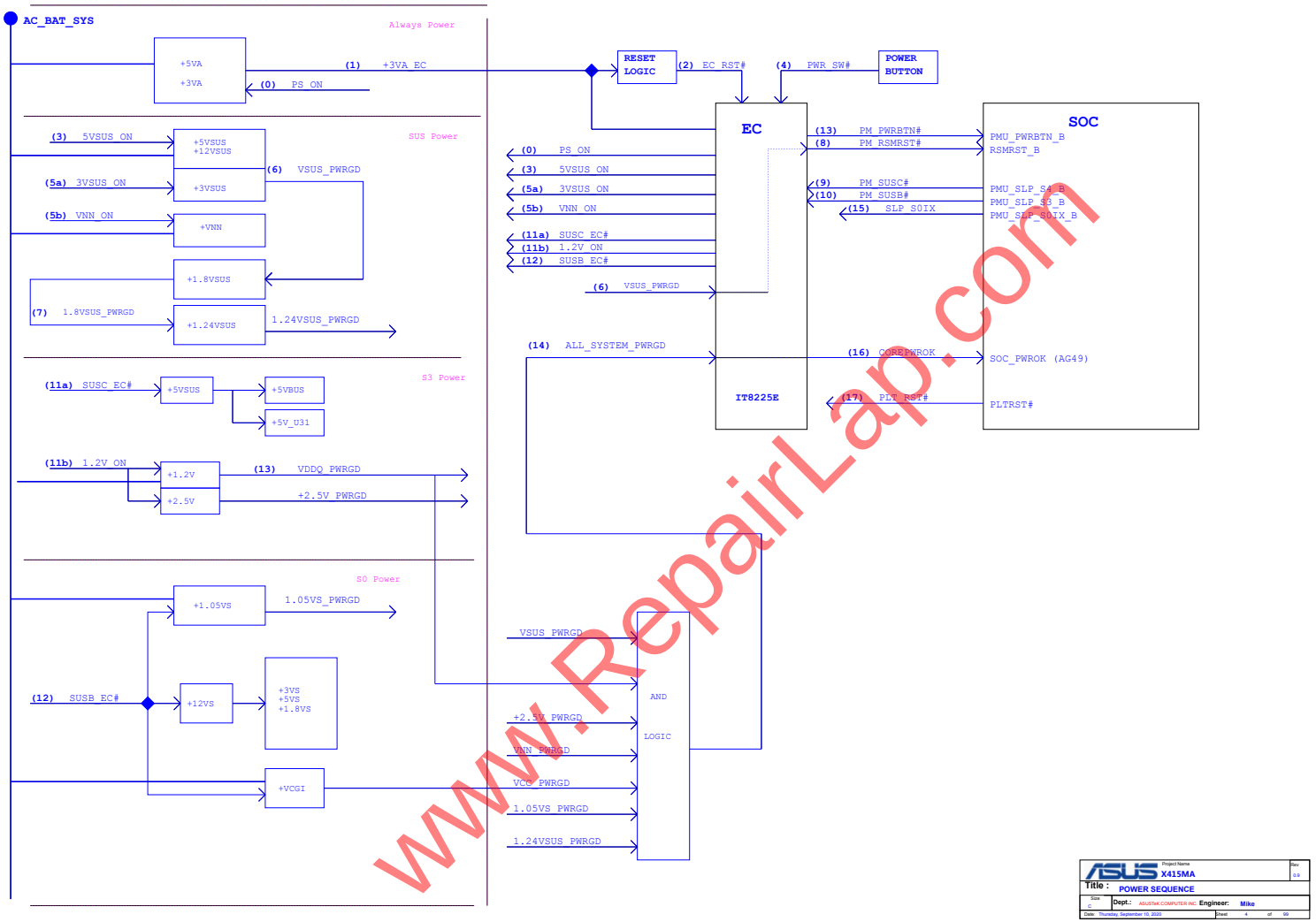
AC-IN Mode

X415MA Power On Sequence Diagram Rev.0.1



ASUS	Project Name: X415MA		Rev: 0.9
Title : POWER SEQUENCE			
Size: C	Dept.: ASUSTeK COMPUTER INC.	Engineer: Mike	
Date: Thursday, September 10, 2020	Sheet 1 of 99		

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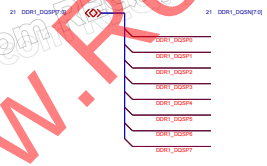
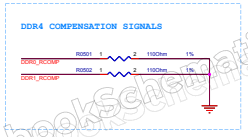
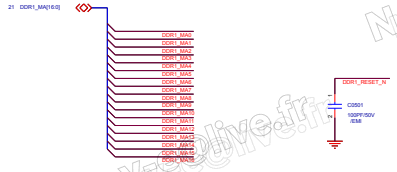
DDR4

Memory Channel A

Memory Channel B







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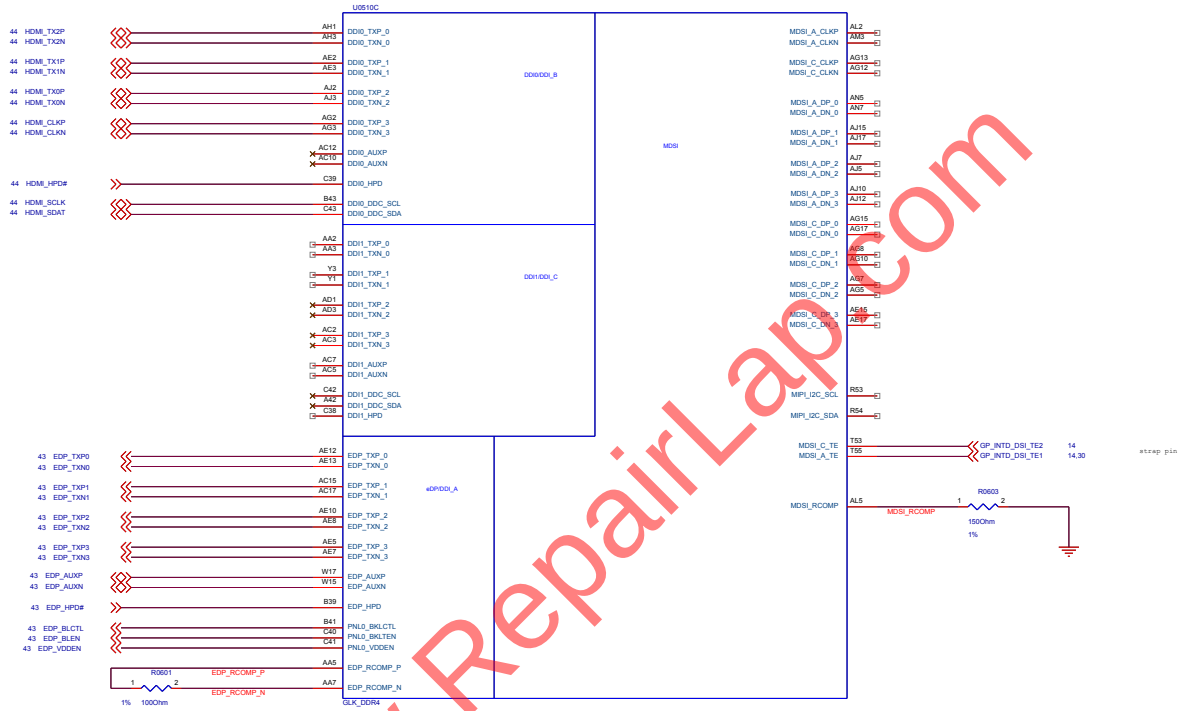
DDI0: HDMI

EDP



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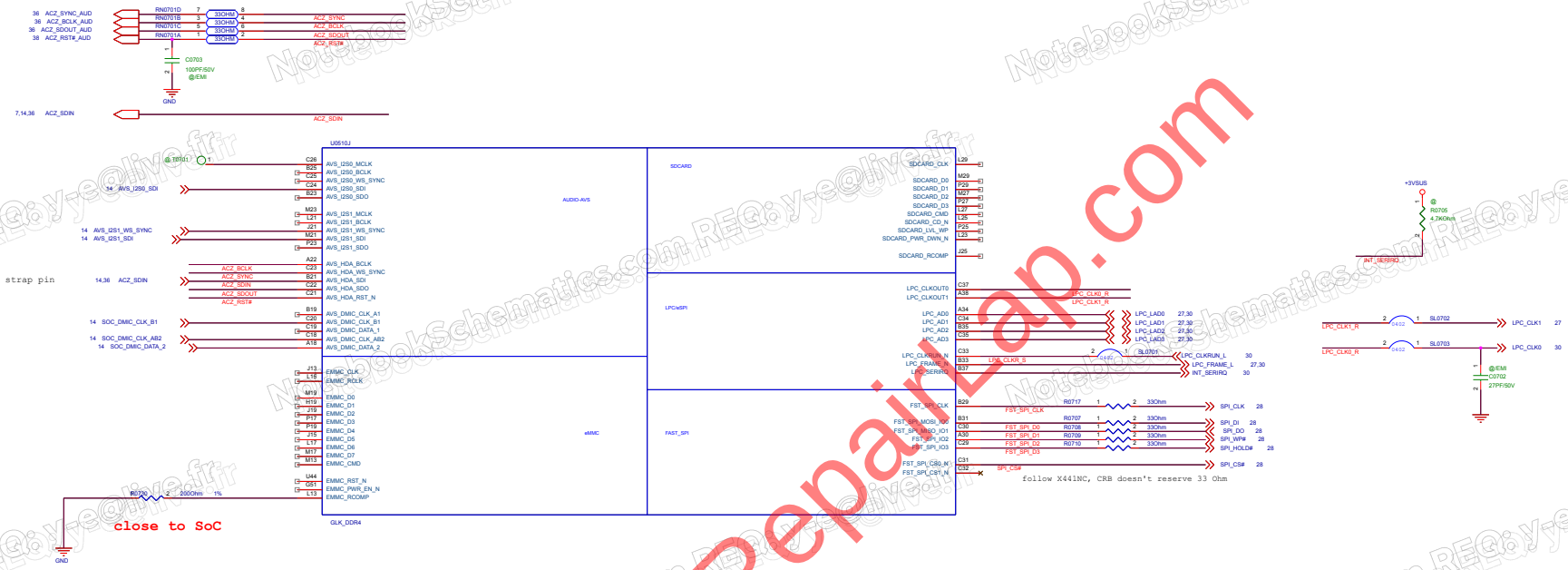

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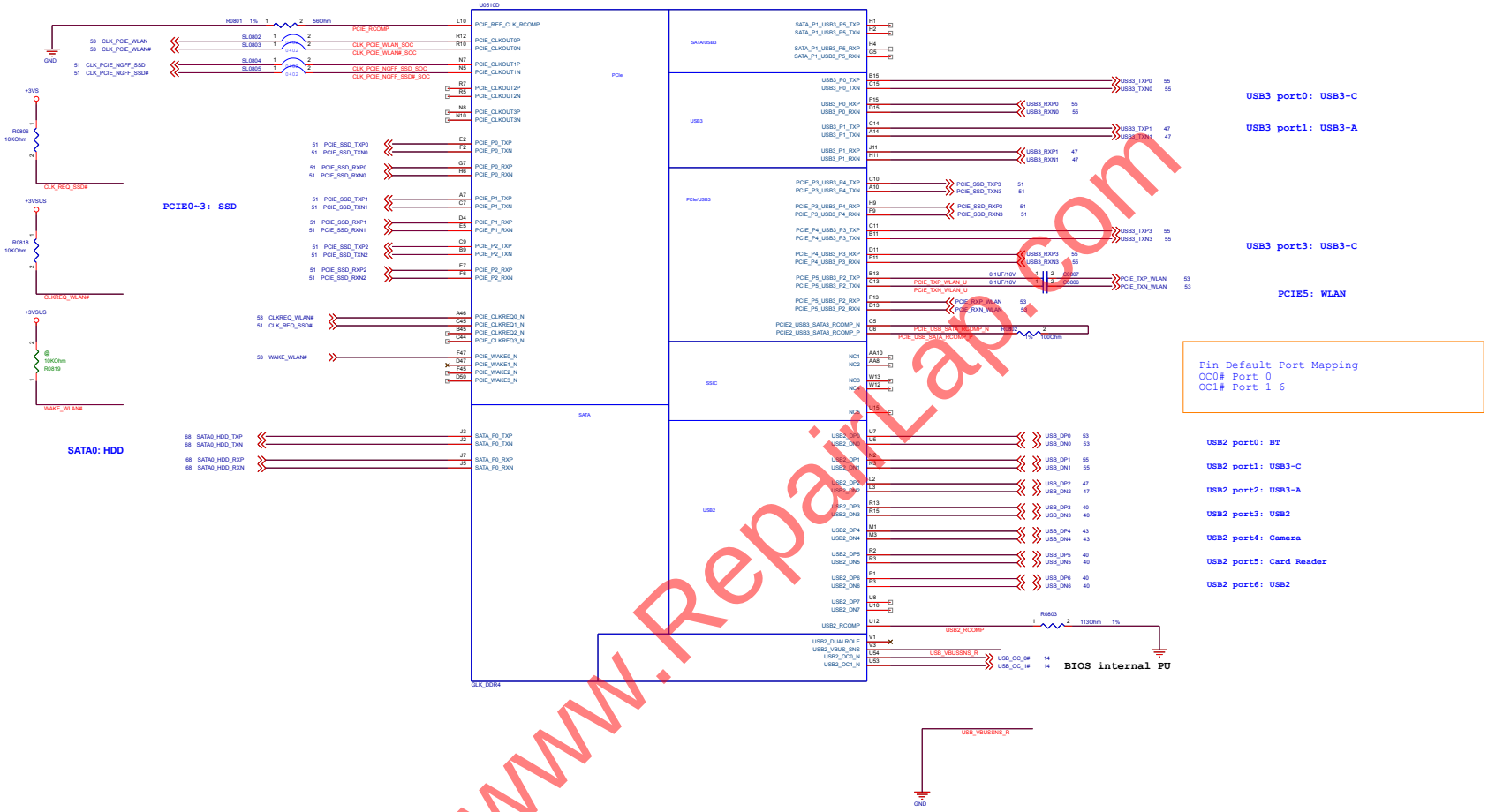

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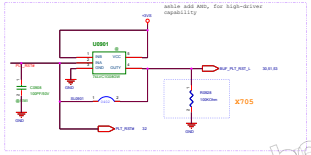
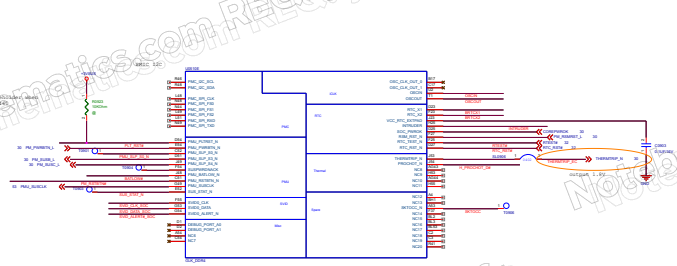
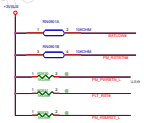
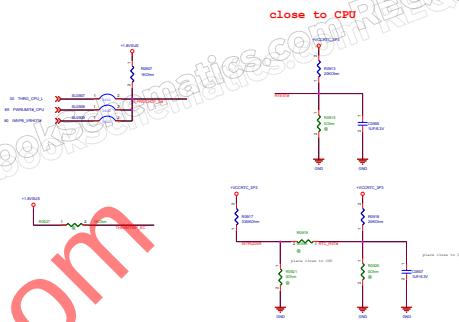
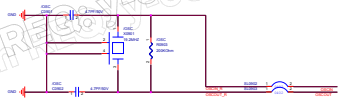


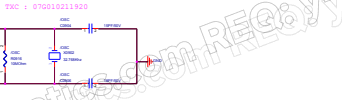


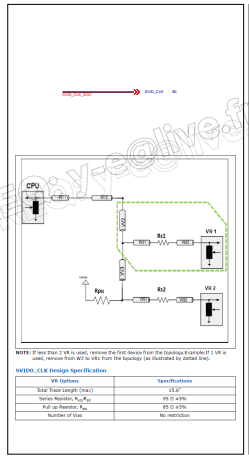




Notes:
1. PMC signals are 1.8V/3.3V tolerant.
2. The I/O voltage selection is done by using Hardware Strap SPI0_IO8. Mode Select: 1=Buffer set to 1.8V mode (default); 0=Buffer set to 3.3V mode.
3. PRCHOT_N and THERMTRIP_N signals are 1.8V tolerant.
4. PMC_SLP_S0_N, PMC_SLP_S3_N, PMC_SLP_S4_N, PMC_SLP_S5_N, PMC_SLP_S3_N and PMC_SLP_S4_N signals. These signals have internal terminations and will suffice for the proper functionality.
5. If the SOC_PWROK signal is connected to the EC then in the S3 state the EC is turned off and there is no termination for the SOC_PWROK signal. Floating SOC_PWROK signal in S3 state can cause high leakage in the VCC 3.3V rail. To address this, make sure there is a weak pull down (~100kohm) path to GND for SOC_PWROK in S3 state to ensure low leakage on RTCAUTOK is indication to SoC on state of power source to System.
6. In adapter mode BATLOW will always be high irrespective of state of charge of battery.In battery mode based on state of charge battery, EC will pull this pin to shutdown SOC.
7. On Gemini Lake platform, the PMC signals such as PMC_SLP_S4_N and PMC_PWROK_N may have small glitches during transition from S0 to S3 and S3 to S0 states. These glitches may cause the SoC to go to unknown state for PMC_SLP_S4_N signals, if happening before RSM_RST_N goes high, are expected behavior on Gemini Lake platform and can be ignored. The glitch on PMC_PWROK_N signal during its assertion can also be ignored if it happens before VCCPGA ramp up. There are no known functional issues as a result of these glitches.



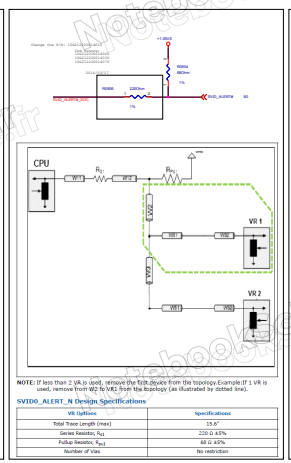




NOTE: If less than 2 VR is used, remove the first device from the topology.Example:If 1 VR is used, remove from W2 to VR1 from the topology (as illustrated by dotted line).

SVID0_CLK Design Specification

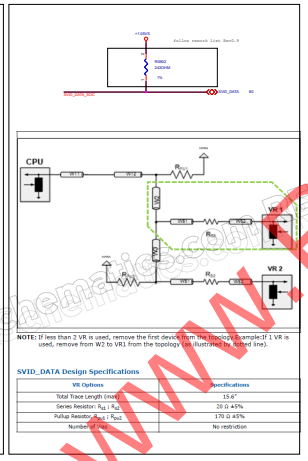
VR Options	Specifications
Total Trace Length (max)	15.6"
Series Resistor, R_{Rs1},R_{Rs2}	95 Ω ±5%
Pull up Resistor, R_{pu}	85 Ω ±5%
Number of Vias	No restriction



NOTE: If less than 2 VR is used, remove the first device from the topology.Example:If 1 VR is used, remove from W2 to VR1 from the topology (as illustrated by dotted line).

SVID0_ALERT_N Design Specifications

VR Options	Specifications
Total Trace Length (max)	15.6"
Series Resistor, R_{R1}	220 Ω ±5%
Pullup Resistor, R_{PU1}	68 Ω ±5%
Number of Vias	No restriction



NOTE: If less than 2 VR is used, remove the first device from the topology.Example:If 1 VR is used, remove from W2 to VR1 from the topology (as illustrated by dotted line).

SVID_DATA Design Specifications

VR Options	Specifications
Total Trace Length (max)	15.6"
Series Resistor, R_{R1} / R_{R2}	20 Ω ±5%
Pullup Resistor, R_{PU1} / R_{PU2}	170 Ω ±5%
Number of Vias	No restriction

ASUS	Project Name: X415MA	Rev: 1.0
Title : Block Diagram		
Size:	Dept.:	Engineer: Mike
Date: Tuesday, September 10, 2019	Sheet 1 of 46	

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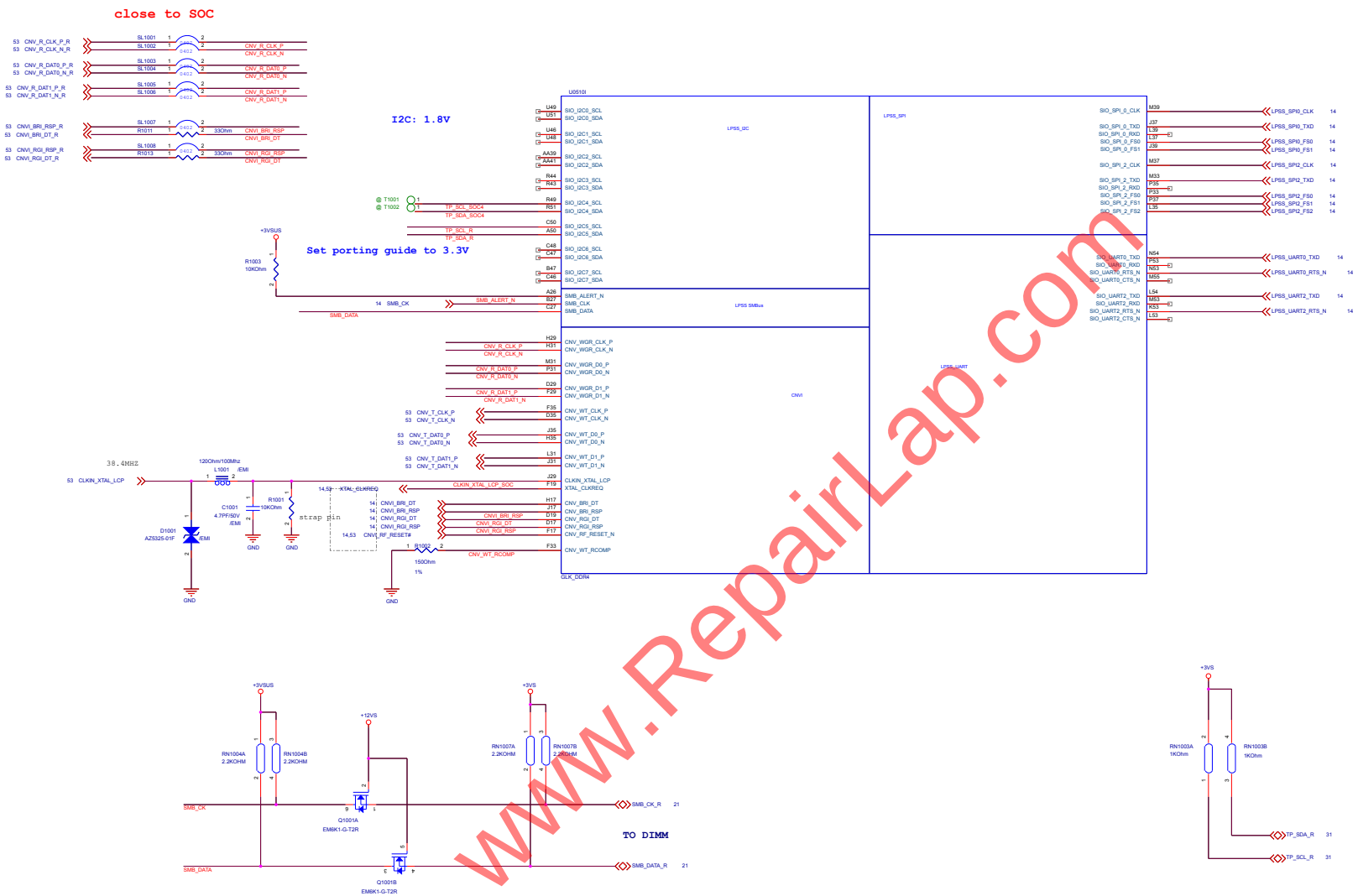

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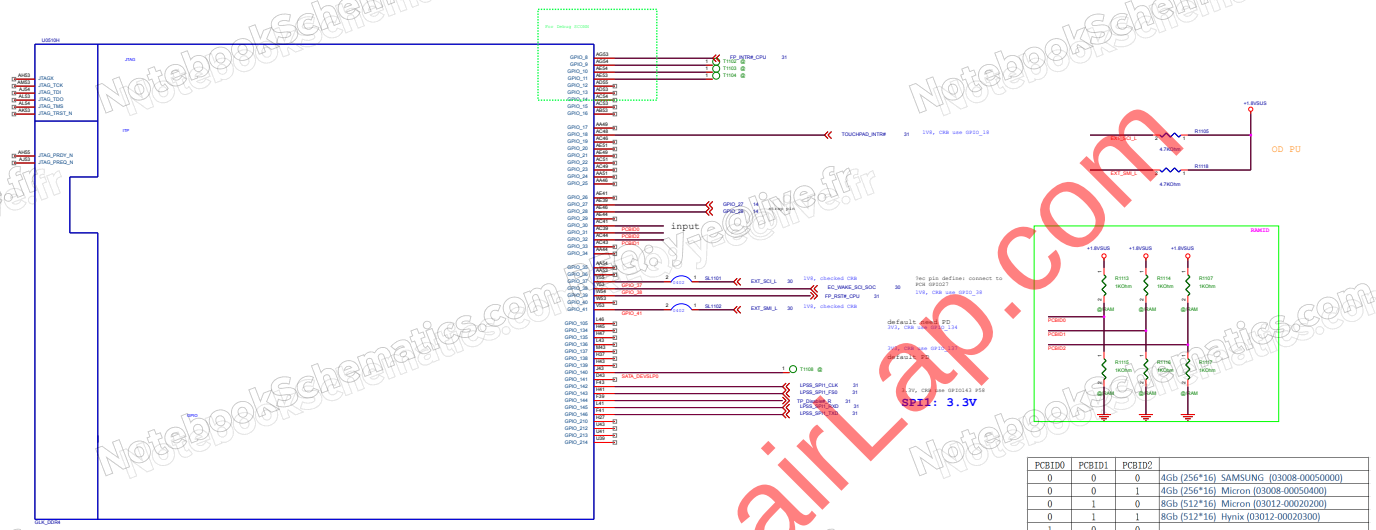

PCBID0	PCBID1	PCBID2	
0	0	0	4Gb (256*16) SAMSUNG (03008-00050000)
0	0	1	4Gb (256*16) Micron (03008-00050400)
0	1	0	8Gb (512*16) Micron (03012-00020200)
0	1	1	8Gb (512*16) Hynix (03012-00020300)
1	0	0	
1	0	1	
1	1	0	
1	1	1	

ASUS	Project Name: X415MA	Rev: 0.4
Title : CPU_POWER		
Size: Custom	Dept.: ASUSTeK COMPUTER INC.	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 11 of 98	

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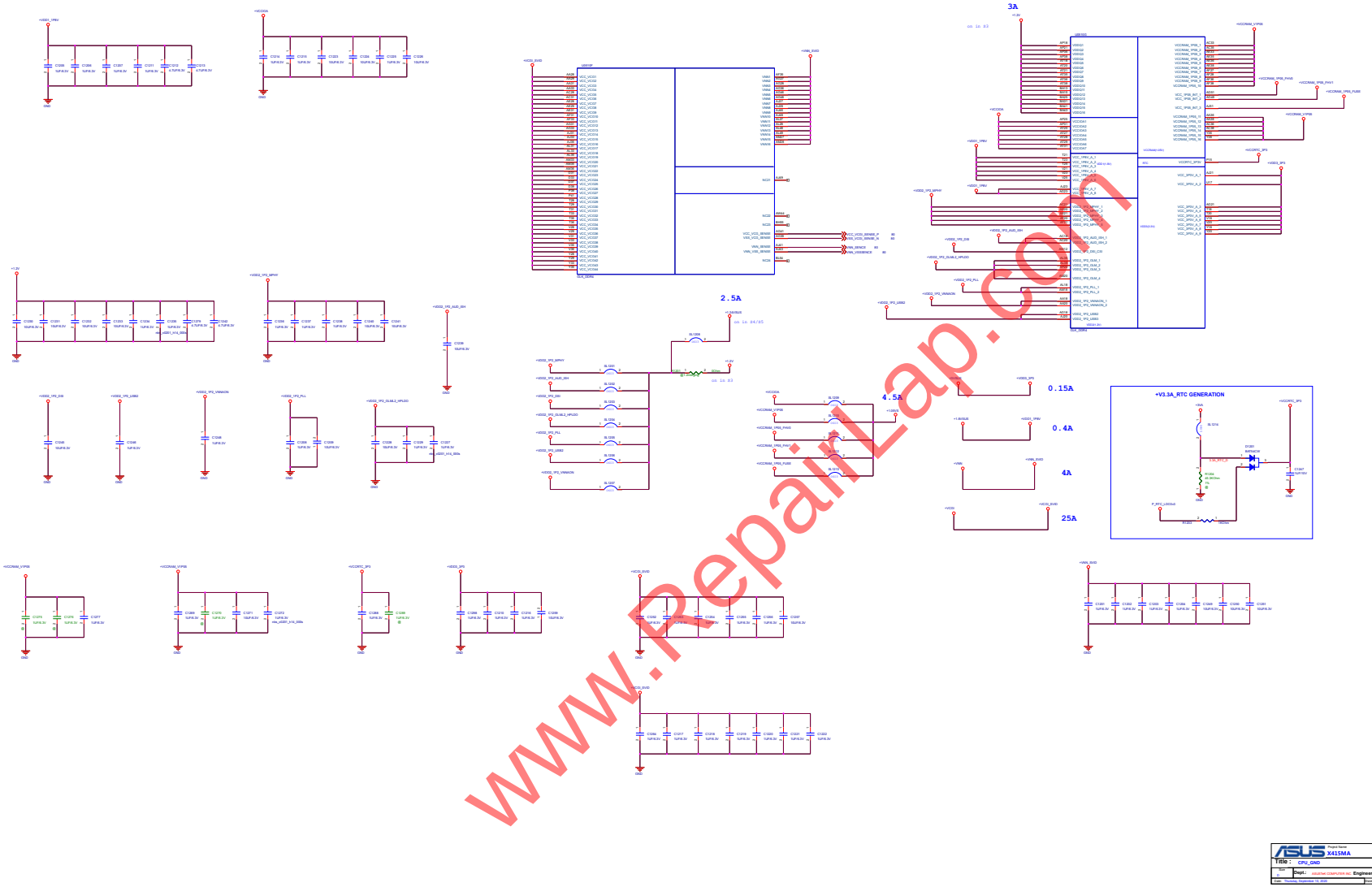

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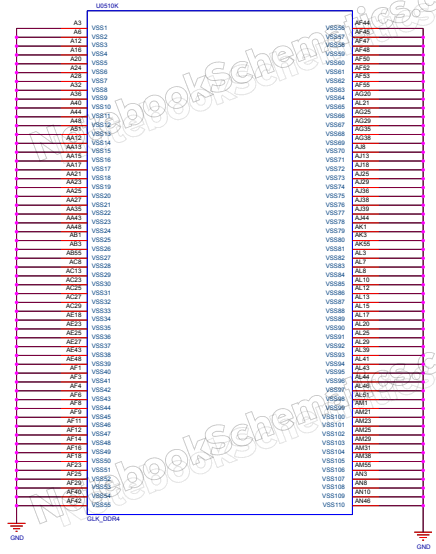

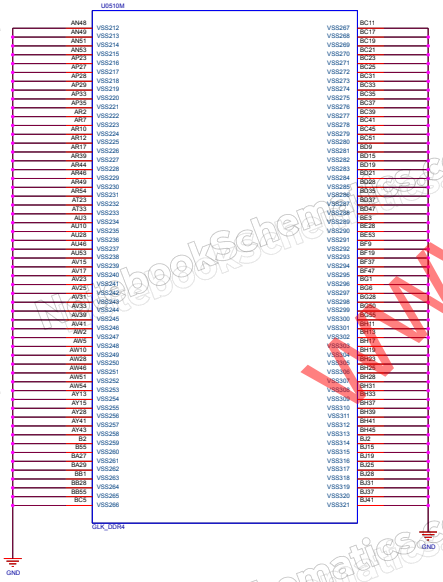

ASUS	Project Name	Rev
	X415MA X515MA	0.9

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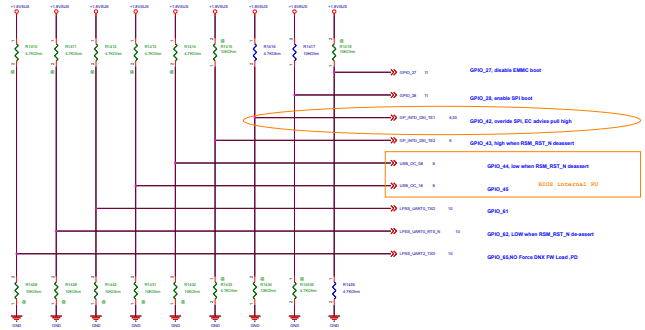

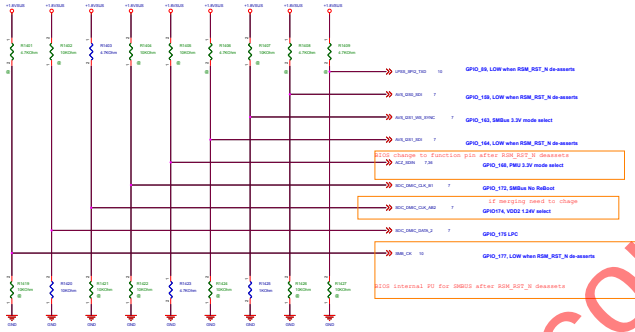

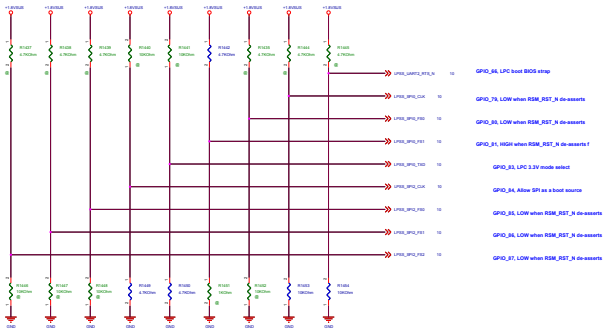

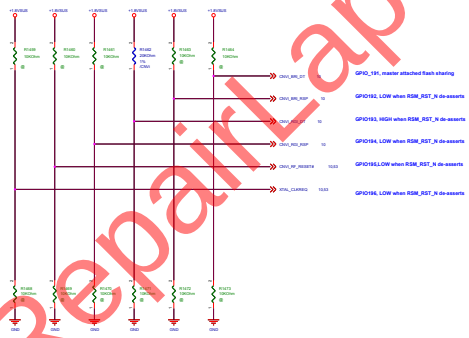

Table 2-28. Hardware Straps (Sheet 1 of 3)

GPIO #	Pin Name	Purpose	Internal Termination	Pin Strap Usage/Description/Polarity
GPIO_27	GPIO_27	Allow eMMC as a boot source	20K PU	1=enable (default); 0=disable; If platform is using SPI as the boot device, then provide a pull-down for this strap to disable eMMC
GPIO_28	GPIO_28	Allow SPI as a boot source	20K PU	1=enable (default) 0=disable ***Note:*** If platform is using eMMC as boot device, then provide a pull down for this strap to disable SPI.
GPIO_42	MDSI_A_TE	Flash Descriptor Override	20K PD	0 = No Override (Normal Operation) 1 = Override ***Note:*** This strap enables the platform to override security features in the SPI.
GPIO_43	MDSI_C_TE	RSVD	20K PU	Ensure that this strap is pulled HIGH when RSM_RST_N de-asserts for normal platform operation.
GPIO_44	USB2_OC0_N	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_45	USB2_OC1_N	Top swap override	20K PD	1 = Enable 0 = Disable (default) ***Note:*** Within the SPI ROM there may be different locations where the boot code is stored. This strap enables platform to change where the core will look for BIOS code for a SPI boot only.
GPIO_61	SIO_UART0_TXD	Enable TXE ROM Bypass	20K PD	1 = enable bypass 0 = disable bypass (default) ***Note:*** This strap tells TXE 3.0 to bypass Read-Only Memory (ROM) that it has on SoC. If an issue occurs with the boot up code of TXE3.0 before the first patch point this strap enabled the platform tell TXE 3.0 to bypass the ROM causing the issue and go to the patch space instead.
GPIO_62	SIO_UART0_RTS_N	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.

follow default

Table 2-28. Hardware Straps (Sheet 2 of 3)

GPIO #	Pin Name	Purpose	Internal Termination	Pin Strap Usage/Description/Polarity
GPIO_65	SIO_UART2_TXD	Force DNX FW Load	20K PD	1 = Force 0 = Do not force (default) ***Notes:*** 1. DnX: Download and Execute 2. This strap is a recovery strap for corrupted FW image. This strap will force TXEI3.0 to execute a "Download and Execute" (DnX) flow, where it would download a new firmware image from a recovery host, over USB, and overwrite the image in the storage media.TXE can do it for BIOS part of FW, but if TXE FW itself is corrupted we need this strap.
GPIO_66	SIO_UART2_RTS_N	LPC boot BIOS strap	20K PD	1=boot from LPC; 0=do not boot from LPC (default) ***Note:*** The board should strap this low and do not use otherwise
GPIO_79	SIO_SPI_0_CLK	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_80	SIO_SPI_0_FS0	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_81	SIO_SPI_0_FS1	RSVD	20K PU	Ensure that this strap is pulled HIGH when RSM_RST_N de-asserts for normal platform operation.
GPIO_83	SIO_SPI_0_TXD	LPC 1.8V/3.3V mode select	20K PD	1=buffers set to 1.8V mode 0=buffers set to 3.3V mode (default)
GPIO_84	SIO_SPI_2_CLK	Allow SPI as a boot source	20K PU	1=disable 0=enable (default)
GPIO_85	SIO_SPI_2_FS0	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_86	SIO_SPI_2_FS1	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_87	SIO_SPI_2_FS2	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_89	SIO_SPI_2_TXD	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_159	AVS_I2S0_SDI	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_163	AVS_I2S1_WS_SYNC	SMBus 1.8V/3.3V mode select	20K PD	1=buffers set to 1.8V mode 0=buffers set to 3.3V mode (default)
GPIO_164	AVS_I2S1_SDI	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_168	AVS_HDA_SDI	PMU (Power Management Unit) 1.8V/3.3V mode select	20K PD	1=buffers set to 1.8V mode 0=buffers set to 3.3V mode (default)
GPIO_172	AVS_M_CLK_B1	SMBus No Re-Boot	20K PD	1 = Enable 0 = Disable (default) ***Note:*** Platforms should strap this LOW. Functionality is handled by the PMC.
GPIO_174	AVS_M_CLK_AB2	VDD2 1.24V vs. 1.20V select	20K PD	1=VDD2 is 1.24V; 0=VDD2 is 1.20V (default)

Table 2-28. Hardware Straps (Sheet 3 of 3)

GPIO #	Pin Name	Purpose	Internal Termination	Pin Strap Usage/Description/Polarity
GPIO_175	AVS_M_DATA_2	eSPI vs. LPC	20K PD	1=eSPI mode; 0=LPC mode (default) Note: The default for A0 will be eSPI due to a bug on LPC.
GPIO_177	SMB_CLK	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_191	CNV_BRI_DT	eSPI Flash Sharing Mode	20K PD	eSPI Flash Sharing Mode: 1=slave attached flash sharing (SAFS); 0=master attached flash sharing (MAFS; default) Note: if eSPI mode is disabled (eSPI/LPC hard strap(GPIO_175) is set to select LPC) then the eSPI slave attached flash sharing strap must also be set to 0.
GPIO_192	CNV_BRI_RSP	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_193	CNV_RGI_DT	RSVD	20K PU	Ensure that this strap is pulled HIGH when RSM_RST_N de-asserts for normal platform operation.
GPIO_194	CNV_RGI_RSP	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_195	CNV_RF_RESET_N	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.
GPIO_196	XTAL_CLKREQ	RSVD	20K PD	Ensure that this strap is pulled LOW when RSM_RST_N de-asserts for normal platform operation.

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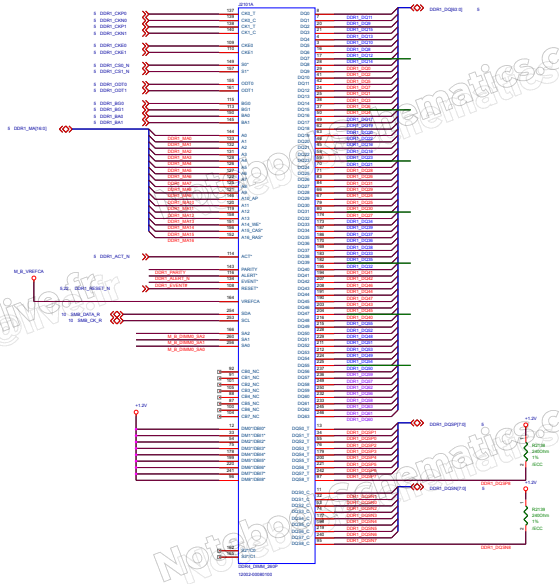
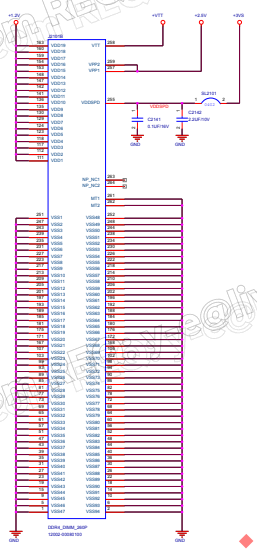
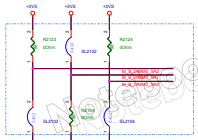
SO DIMM DDR4

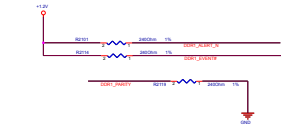




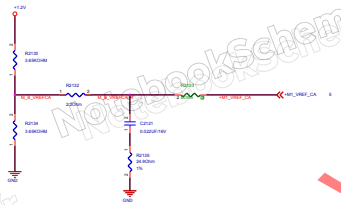


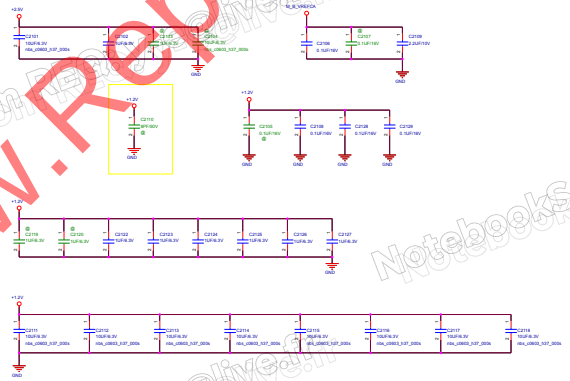
Place close to SO-DIMM

WRITE ADDRESS: 0XA4



All Vref trace must be 15 mils width





Title :	MEM_SODIMM
Engineer:	Mike
Project Name	X415MA

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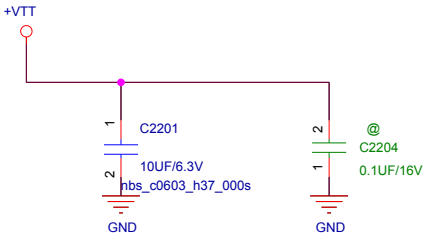

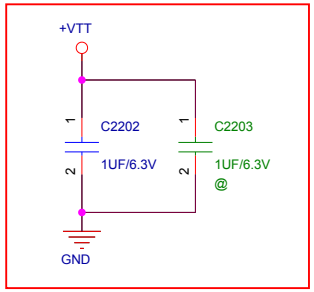

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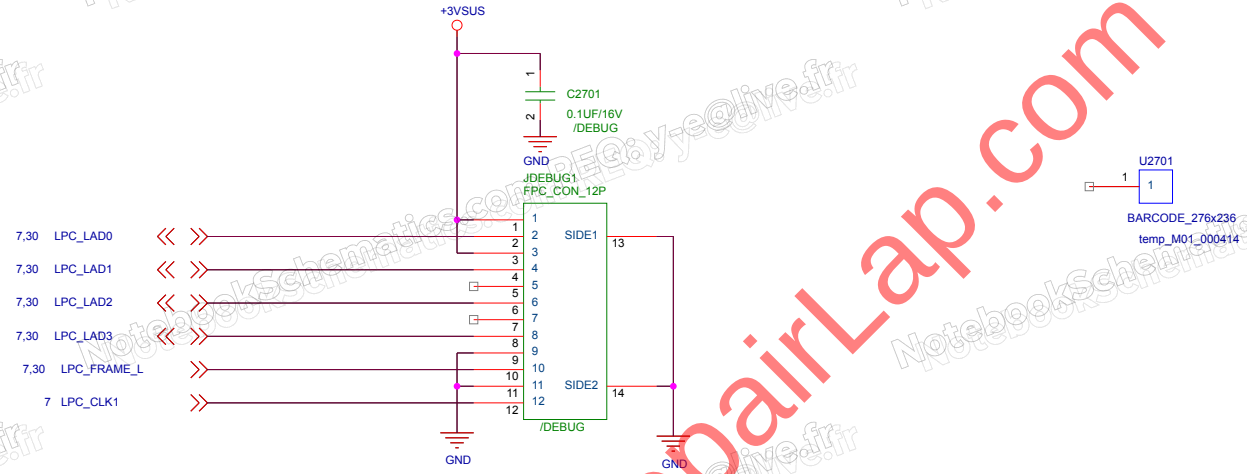
LPC DEBUG PORT



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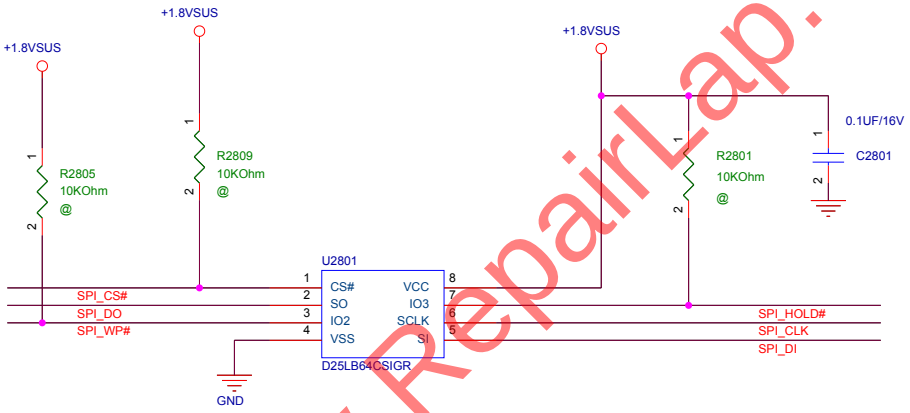
Close to CPU



Q Model料號:
05006-00013500
05006-00014600

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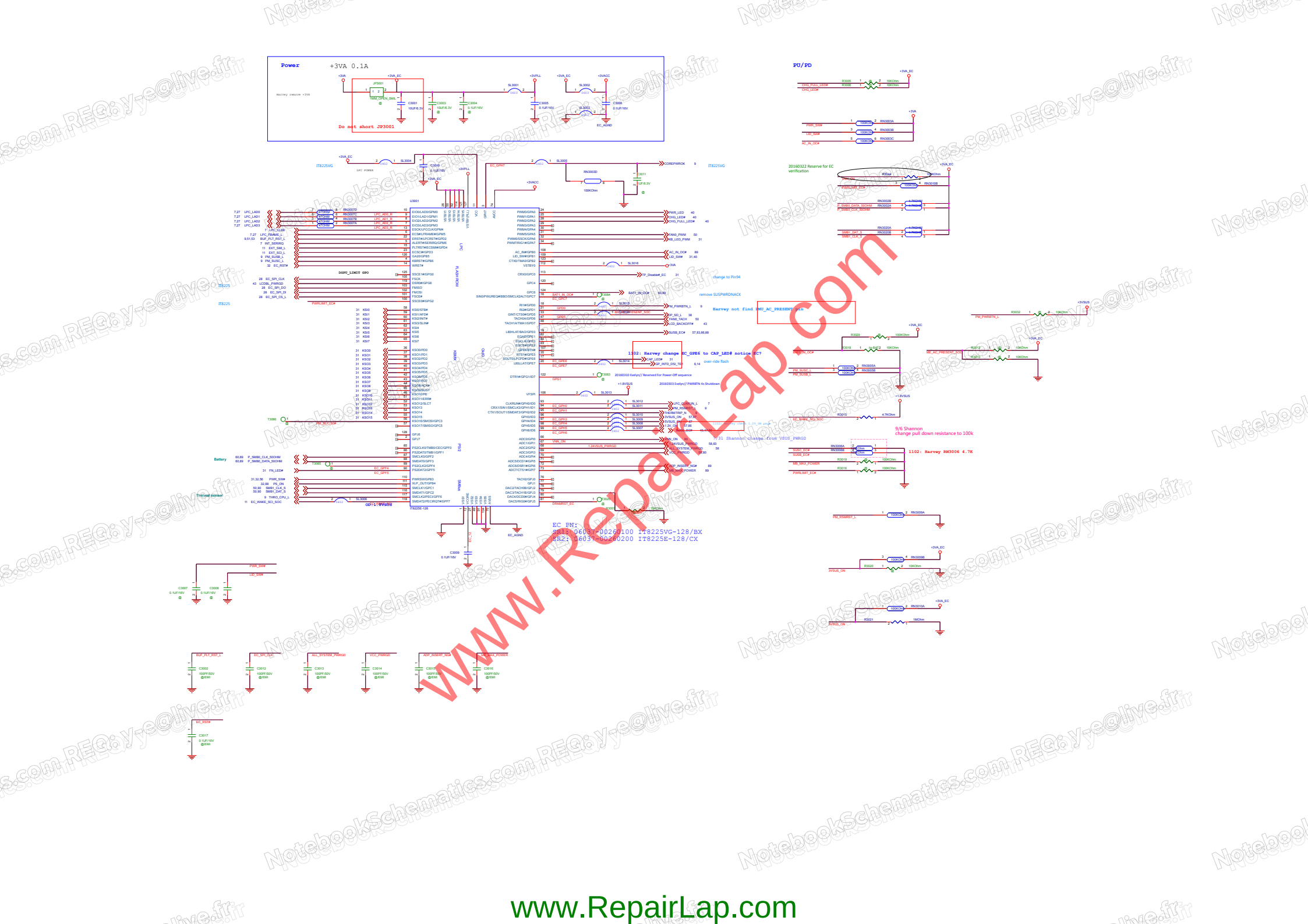
Power
+3VA 0.1A
Do not short JP3001
PU/PD
EC_GPIO7
EC_GPIO6
EC_GPIO5
EC_GPIO4
EC_GPIO3
EC_GPIO2
EC_GPIO1
EC_GPIO0
EC_AGND
Harvey not find IMU_AC_PRESENT_SOC
1102: Harvey change EC_GPE6 to CAP_LED# notice EC?
1102: Harvey RN3006 4.7K
9/6 Shannon
change pull down resistance to 100k
EC PN:
SB1: 06037-00260100 IT8225VG-128/BX
ER2: 06037-00260200 IT8225E-128/CX
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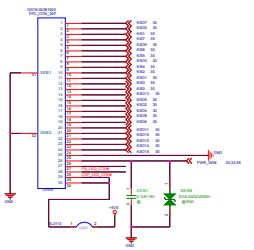
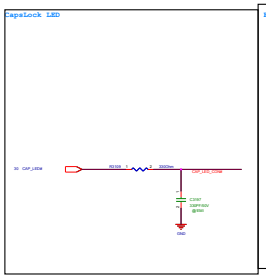
Main Board

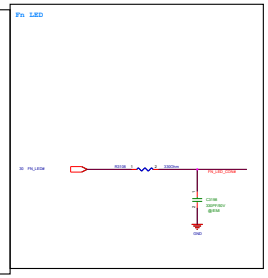
Internal Keyboard



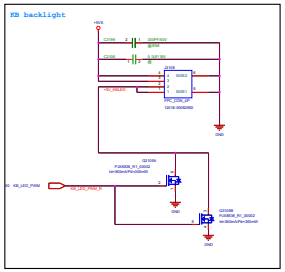
CapsLock LED

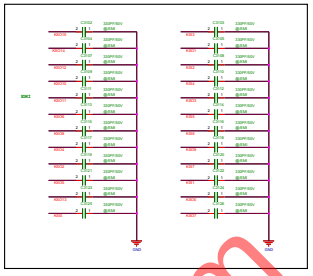


Fn LED

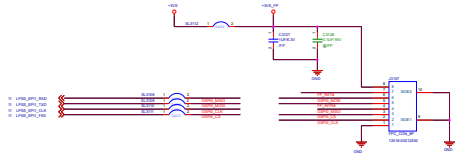


KB backlight

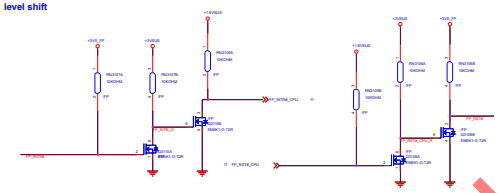




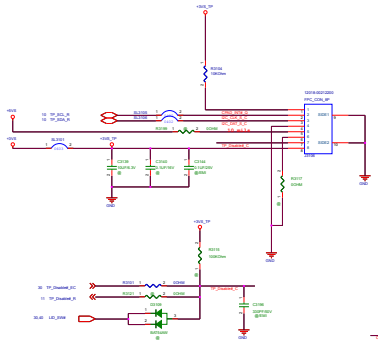
FingerPrint

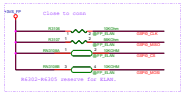


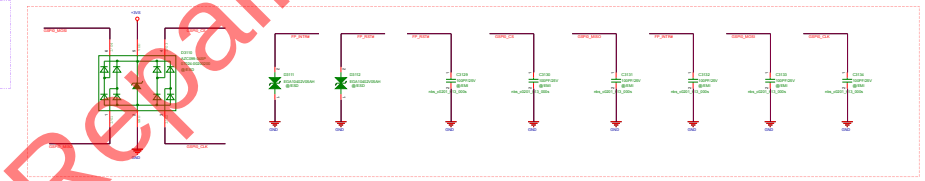
level shift

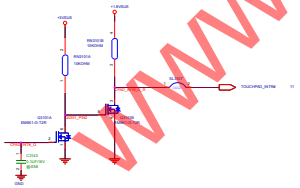


TouchPad



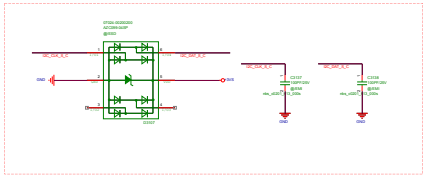






KB Finger printer (Reserved)

PIN NO.	PIN DEFINE
1	VDD_3.3V
2	NC
3	D-
4	D+
5	GND
6	I2C_DATA
7	I2C_CLK
8	TOUCH_INT



CN1		CN2	
PIN NO.	PIN DEFINE	PIN NO.	PIN DEFINE
1	VDD_3.3V_TP	1	VDD_3.3V_FP
2	GND	2	FP_RSTN
3	PS2_CLK	3	FP_MOSI
4	PS2_DATA	4	FP_INTN
5	GND	5	FP_MISO
6	SDA	6	FP_CSN
7	SCL	7	FP_SCK
8	INT	8	GND

Title :	KB & TP
Engineer:	Mike
Project Name	X415MA

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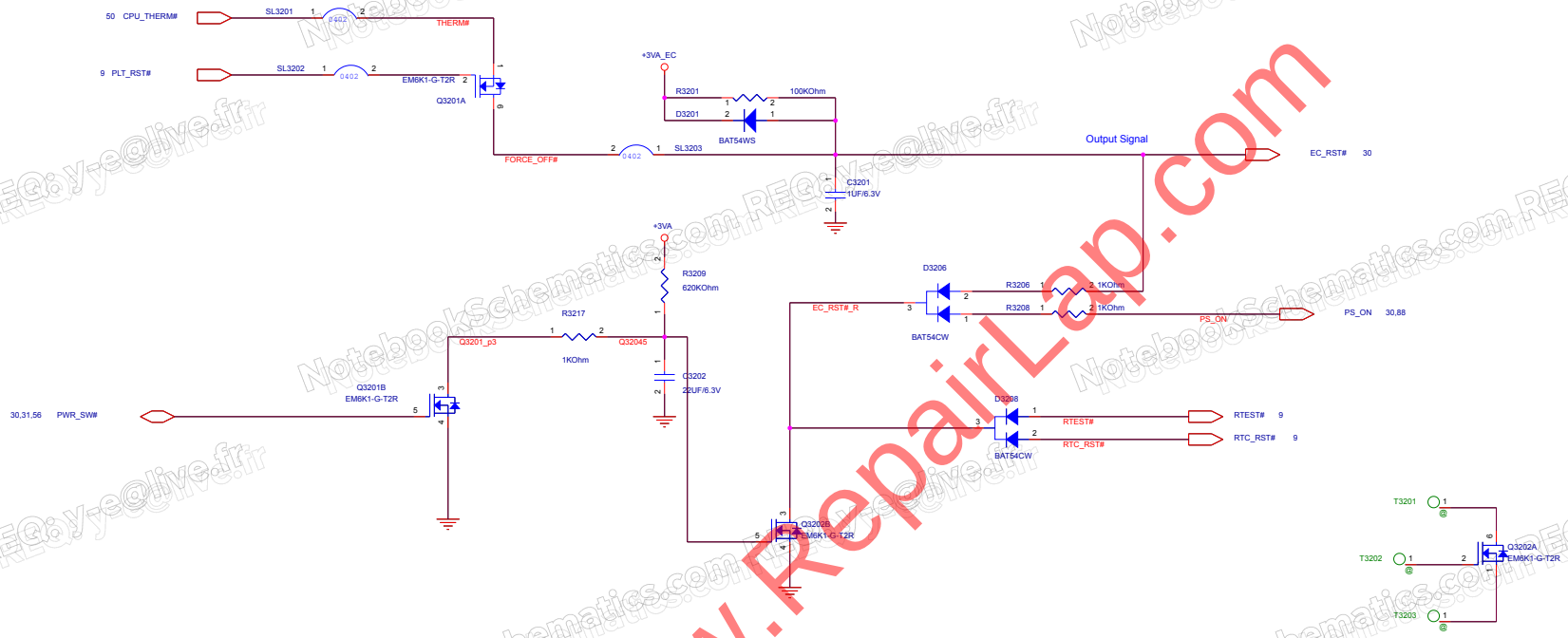
Thermal Policy



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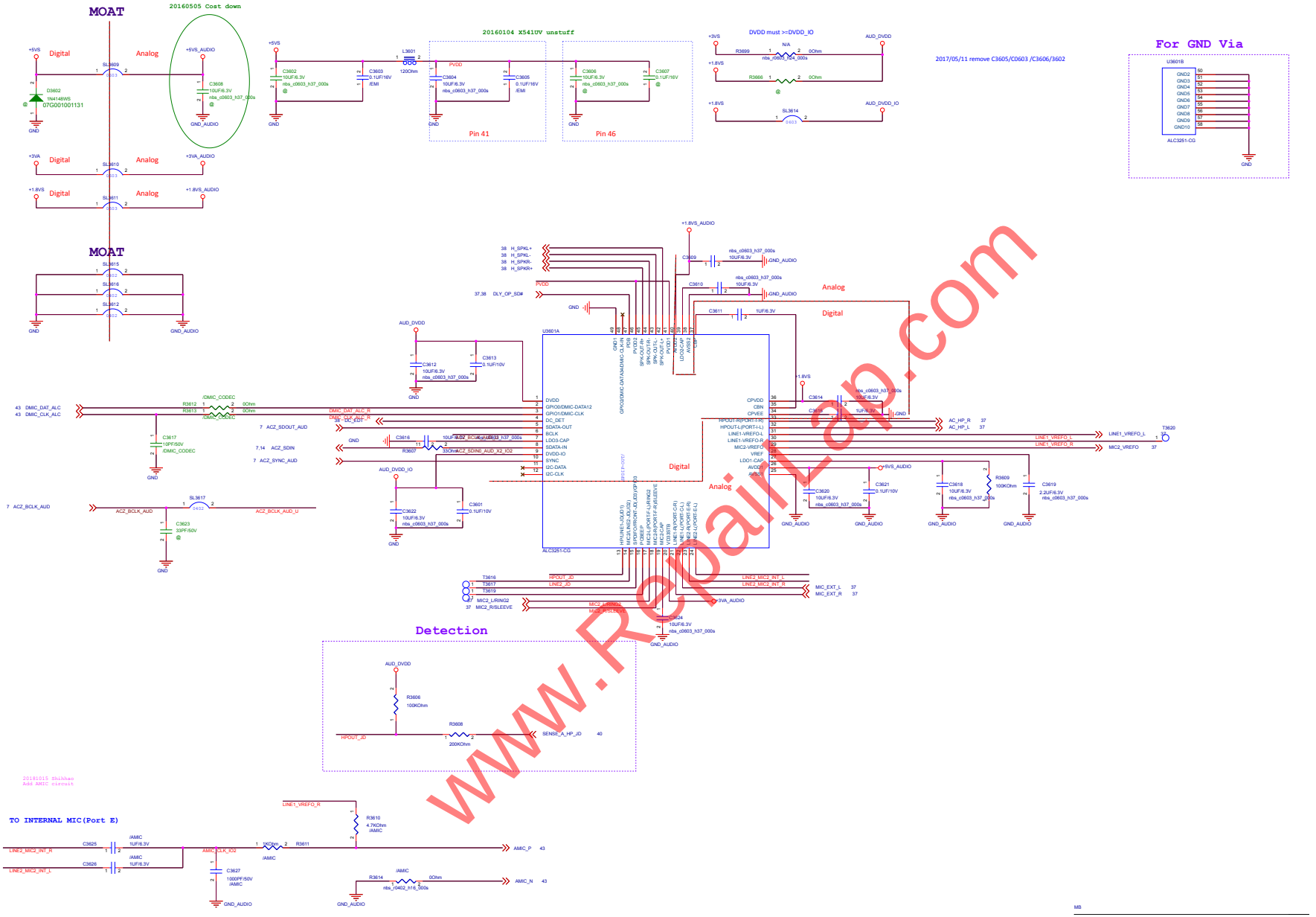

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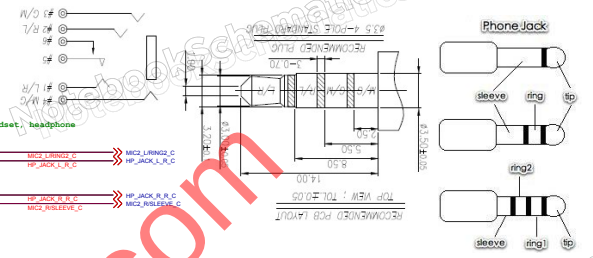
Universal Jack (Normal open type)

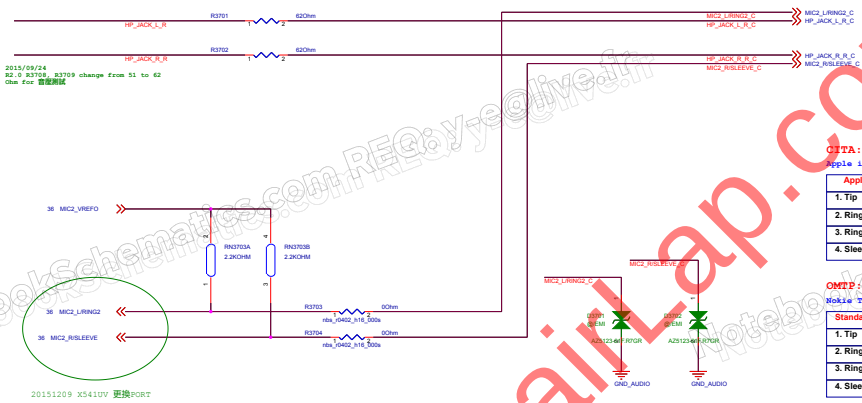
R1.3. Item 12.
Add 4 pole headset jack normal
open type for project demand.

• 尺寸：一般分三種規格，2.5mm、3.5mm、6.35mm。
• 種類：一般分也三種規格，TS 端子 (Mono)、TRS 端子 (Stereo)、TRRS 端子 (Stereo + Mic/Video)。



R1.4. Item 13.
Recommand the HP damping resistance 56 ohm for CB certification.
Depend on your project, you can change the resistance value to meet CB certification
(under 150mV), best chicce is between 140mV~150mV section.

Global Headset
Normal Open
Supported iPhone/Nokia headset, headphone



CTTA：國際標準
Apple iPhone/HTC/小米 的 Phone Jack 定義：

Apple	iPad (Stereo)	iPod (Stereo)	iPhone (Mic)	iPod (AV)
1. Tip	Left channel	Left channel	Left channel	Left channel
2. Ring1	Right channel	Right channel	Right channel	Right channel
3. Ring2	-	-	Ground	Ground
4. Sleeve	Ground	Ground	Mic	Video

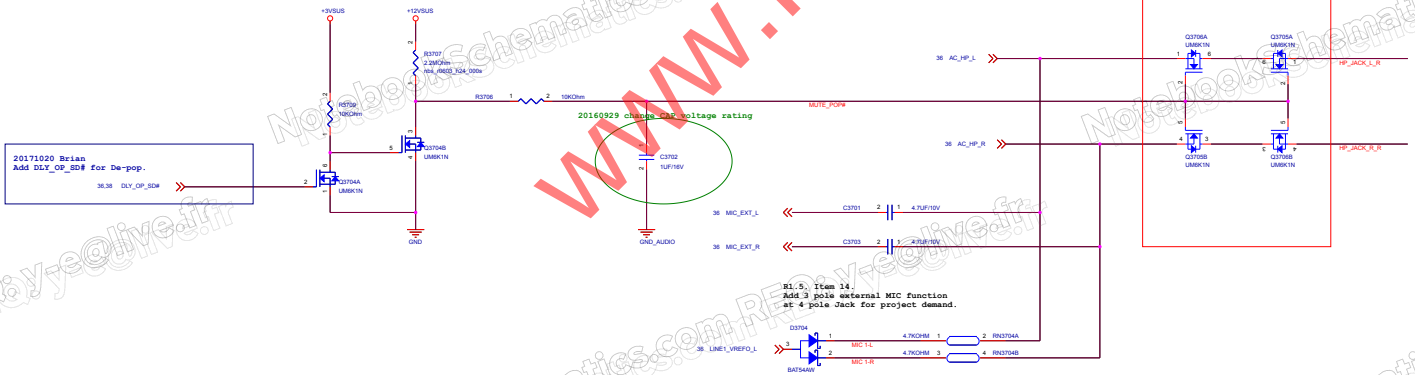
OMTP：國家標準
Nokia Type 的 Phone Jack 定義：

Standard	Mono	Stereo	Stereo + Mic	Audio + Video
1. Tip	Left channel	Left channel	Left channel	Left channel
2. Ring1	-	Right channel	Right channel	Video
3. Ring2	-	-	Mic	Ground
4. Sleeve	Ground	Ground	Ground	Right channel

R0.93. Item 3.
Realtek suggest PCB trace width of RING2 & SLEEVE at least 40 mils.

20180307 Brian
Add D3701/D3702 for EMI MIC burn issue.

耳機pop noise mute線路
MUTE CONTROL



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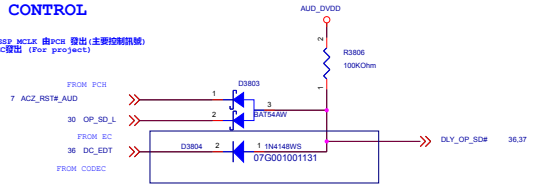
MUTE CONTROL

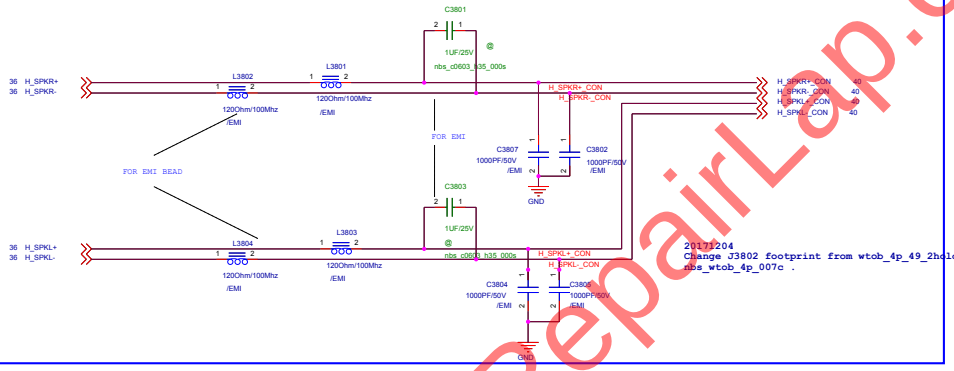
20151216
HDA_RST# SSD_MCLK 由PCH 發出(主要控制訊號)
OP_En 由 EC發出 (For project)



20171020
Change R3833 to D3802 for De-pop.

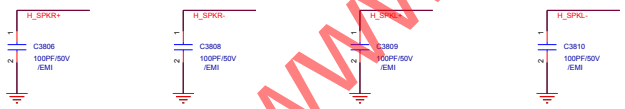
Trace width for
H_SPKL+_O/H_SPKL-_O/H_SPKR+_O/H_SPKR-_OSpeaker
Speaker :4 ohm : 40mil ;8 ohm : 20mil

20180828 Jacky Change C3802,C3803,C3805,C3806 to 1000pF amd mount them



20171204
Change J3802 footprint from wtob_4p_49_2hold_ra_ac to
nbs_wtob_4p_007c .

20170505 不需3W Speaker 改成 short lane



MB

ASUS	Project Name: X415MA	Rev: R09
Title : Speaker		
Size: Custom	Dept.: NB2RD1EE3	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 38 of 99	

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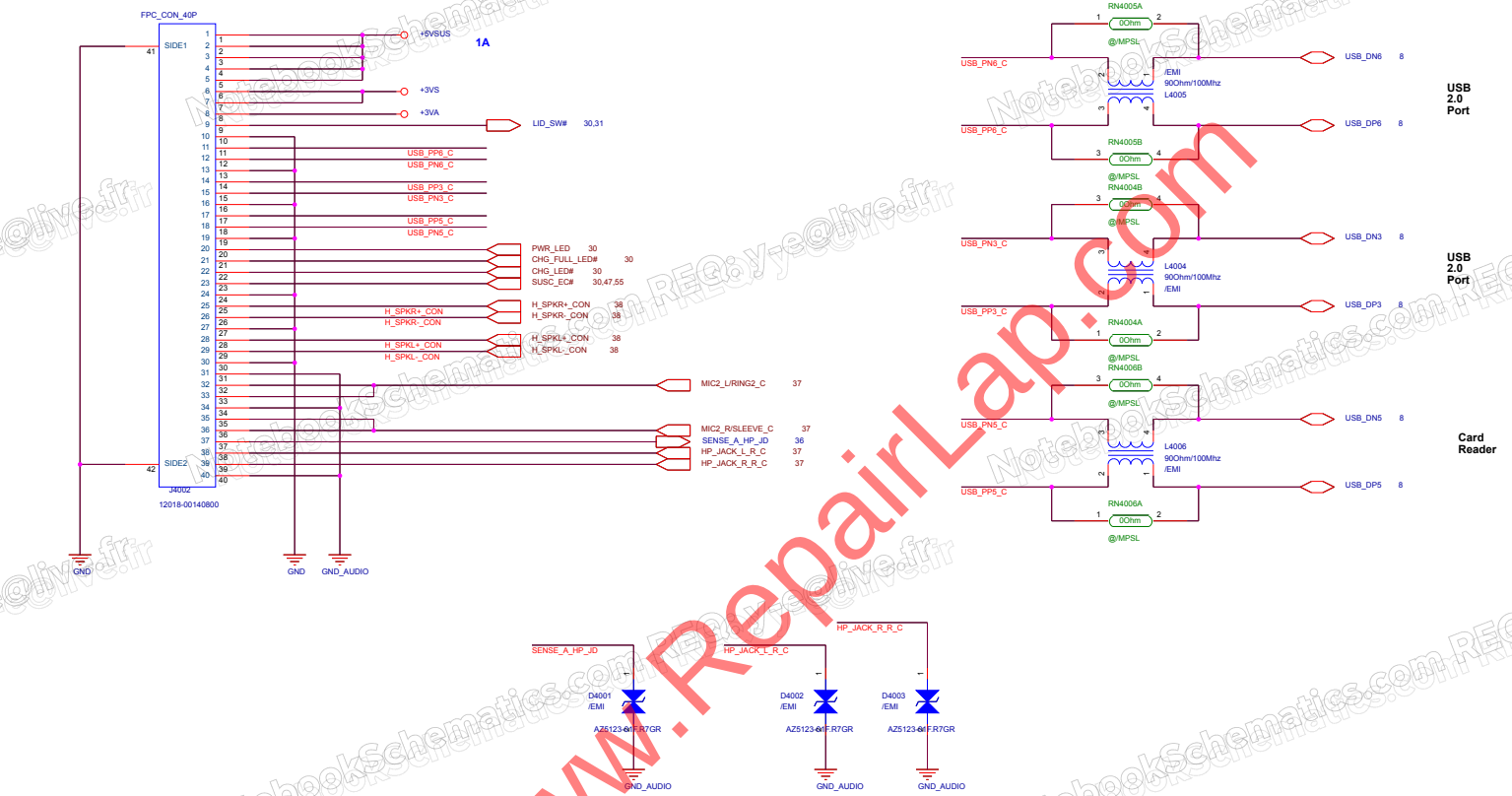
MB to IO



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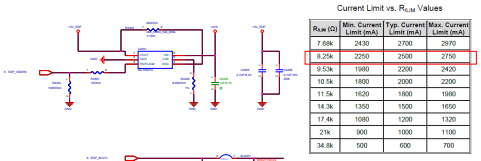
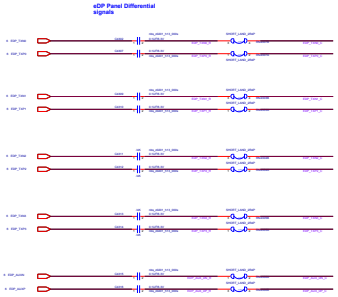
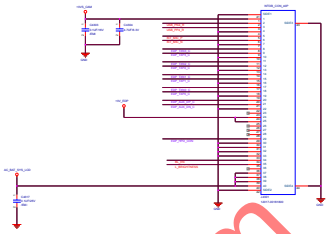
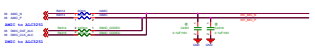


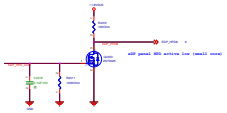
Current Limit vs. R_{ILIM} Values

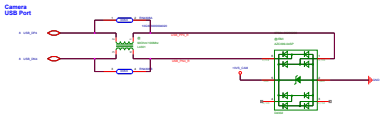
R_{ILIM} (Ω)	Min. Current Limit (mA)	Typ. Current Limit (mA)	Max. Current Limit (mA)
7.68k	2430	2700	2970
8.25k	2250	2500	2750
9.53k	1980	2200	2420
10.5k	1800	2000	2200
11.5k	1620	1800	1980
14.3k	1350	1500	1650
17.4k	1080	1200	1320
21k	900	1000	1100
34.8k	500	600	700

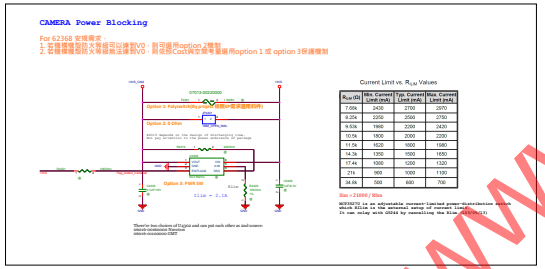


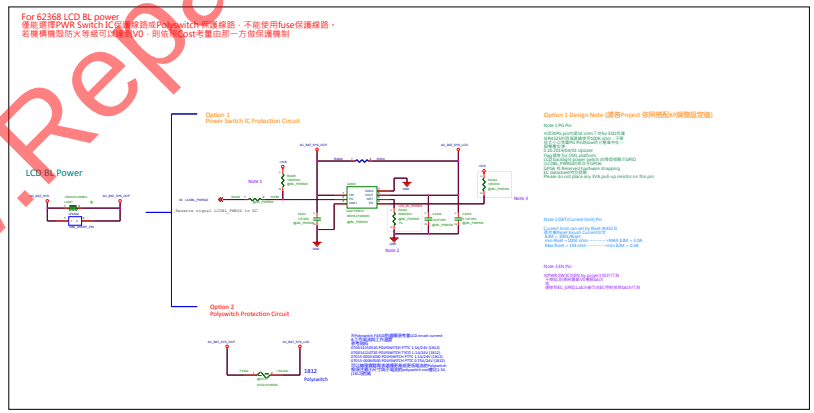












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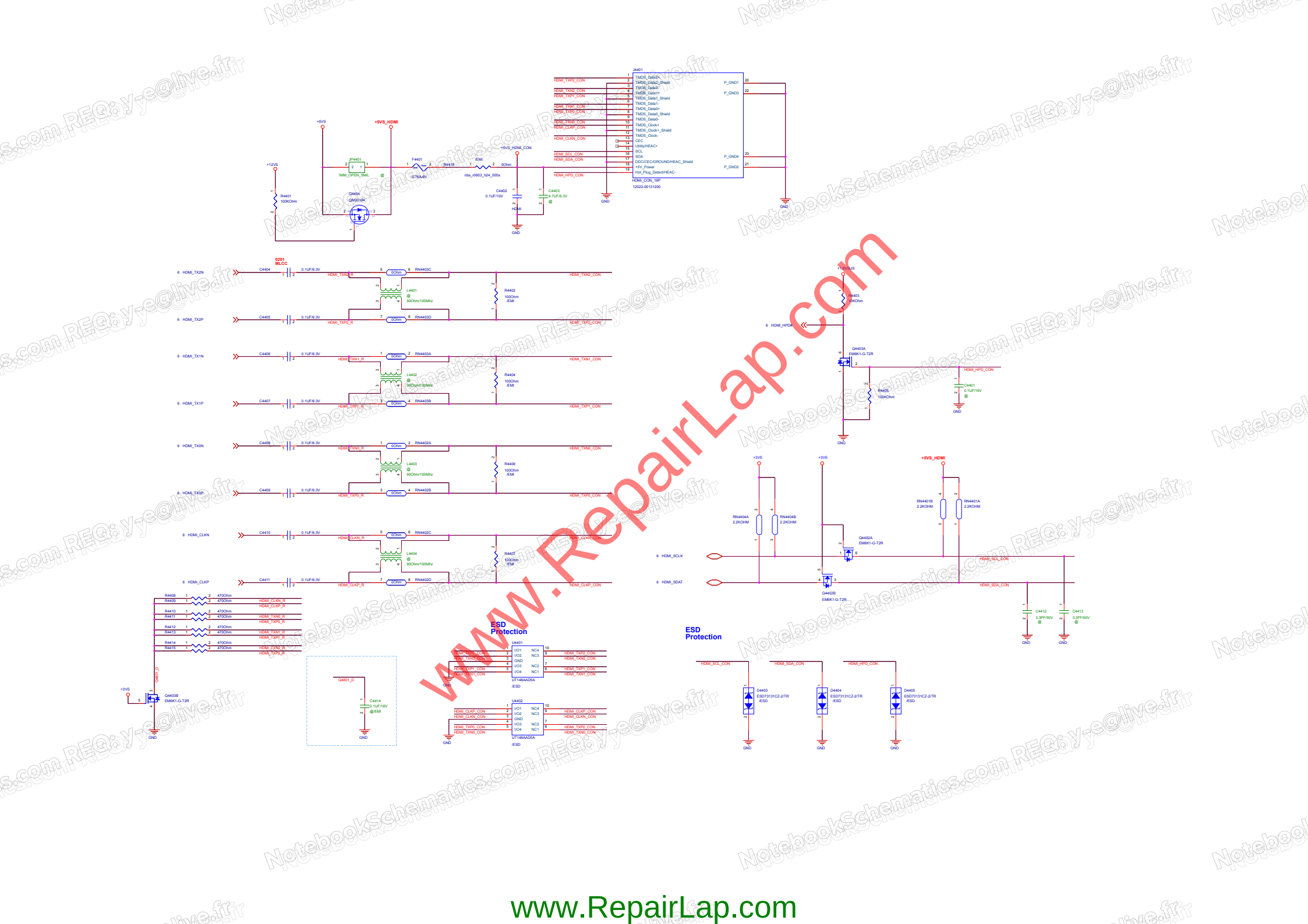

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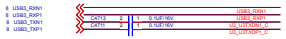
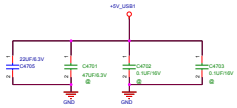
USB3.0 port0

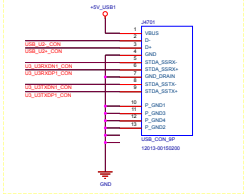


EM PASS 09022-00110000
NPI co-lay
10G3020000040300 RES A 0
OHM(0402)JUMP 2R4P//YAGEO

09022-00110000 PCB footprint option:
EMF mounted -> nbs_filter_6p_001c
RN mounted -> 10G3020000040300

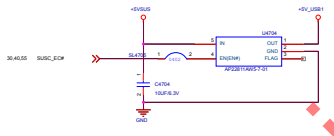


USB3.0 por0 & USB2.0 port2

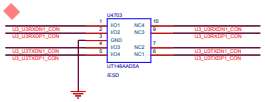


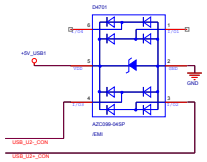
USB2.0 port2

USB3.0 port 0 Power SW for Power Protect



ESD





ESD

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ASUS	Project Name: X415MA	Rev: 0.0
Title : USB Conn		
Size: Custom	Dept.: NBGROHEE3	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 47 of 59	

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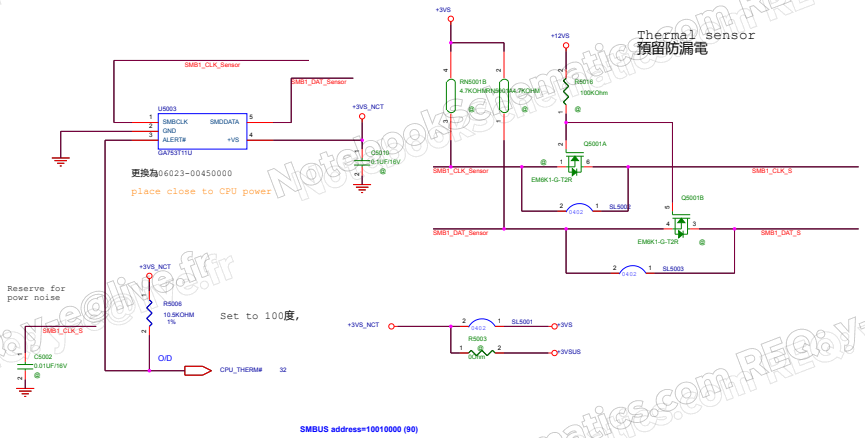
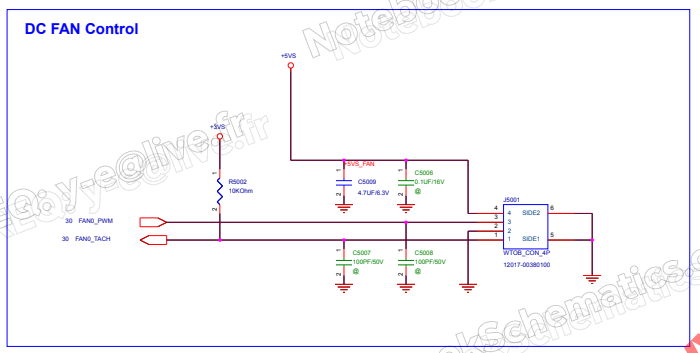
CPU Thermal Sensor

Main Board



DC FAN Control

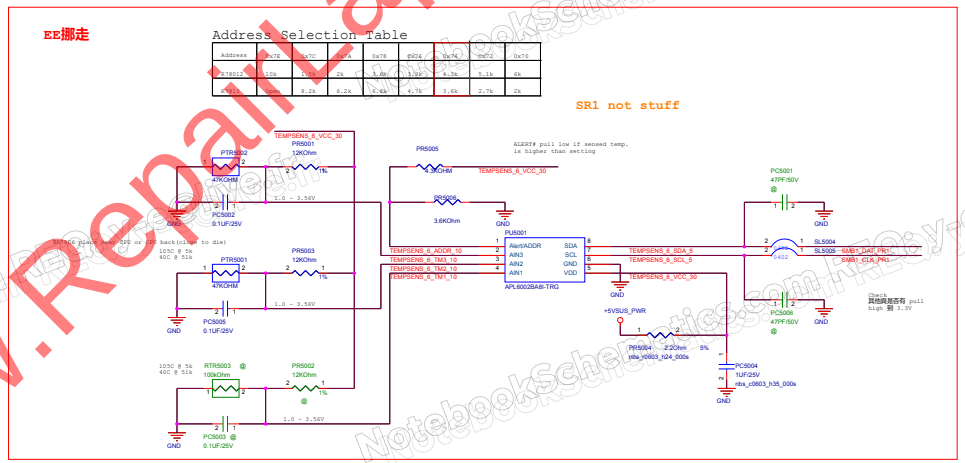


EE挪走

Address Selection Table

Address	0x7E	0x7C	0x7A	0x78	0x76	0x74	0x72	0x70
0x78(1)	12k	10k	8.2k	6.2k	4.99k	3.99k	3.24k	2.49k
0x79	10k	8.2k	6.2k	4.99k	4.7k	3.9k	3.24k	2.49k
0x7A	8.2k	6.2k	4.99k	4.7k	3.9k	3.24k	2.7k	2k
0x7B	6.2k	4.99k	4.7k	3.9k	3.24k	2.7k	2k	1.5k
0x7C	4.99k	4.7k	3.9k	3.24k	2.7k	2k	1.5k	1k
0x7D	4.7k	3.9k	3.24k	2.7k	2k	1.5k	1k	750
0x7E	3.9k	3.24k	2.7k	2k	1.5k	1k	750	500
0x7F	3.24k	2.7k	2k	1.5k	1k	750	500	330

SR1 not stuff



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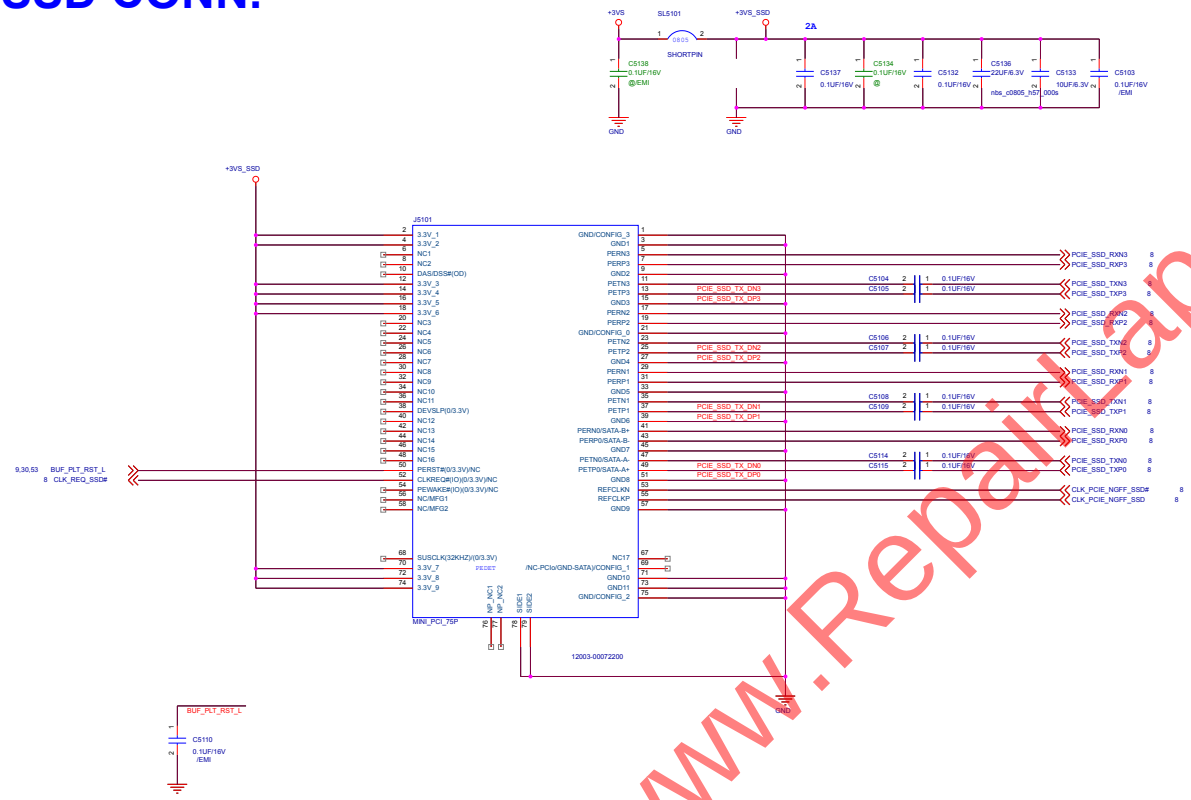
SSD CONN.



36.3.2.4 AC Capacitor General Guidelines for M.2 SSD Storage Routing on SATA / PCI Express* Multiplexed Ports

The following table summarizes the AC capacitor requirements on the motherboard when using the SATA/PCIe* multiplexed ports.

Note: When SATA and PCIe* are muxed, always route according to SATA design guidelines. SATA does not support signal polarity reversal and does not support lane reversal.

Table 36-7. SATA / PCI Express* Gen 2 and Gen 3 Capacitor Values

Condition	PCI Express* Gen 2 Only	PCI Express* Gen 3 Only	SATA Only	PCI Express* Gen 2/ SATA	PCI Express* Gen 3/ SATA
Processor Tx	100 nF	220 nF	10 nF	100 nF	220 nF
Processor Rx	None	None	10 nF[2]	None	None[3]

BOM

ASUS	Project Name: X415MA	Rev: R09
Title : SSD		
Size: Custom	Dept.: NB2RD1EE3	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 51 of 99	

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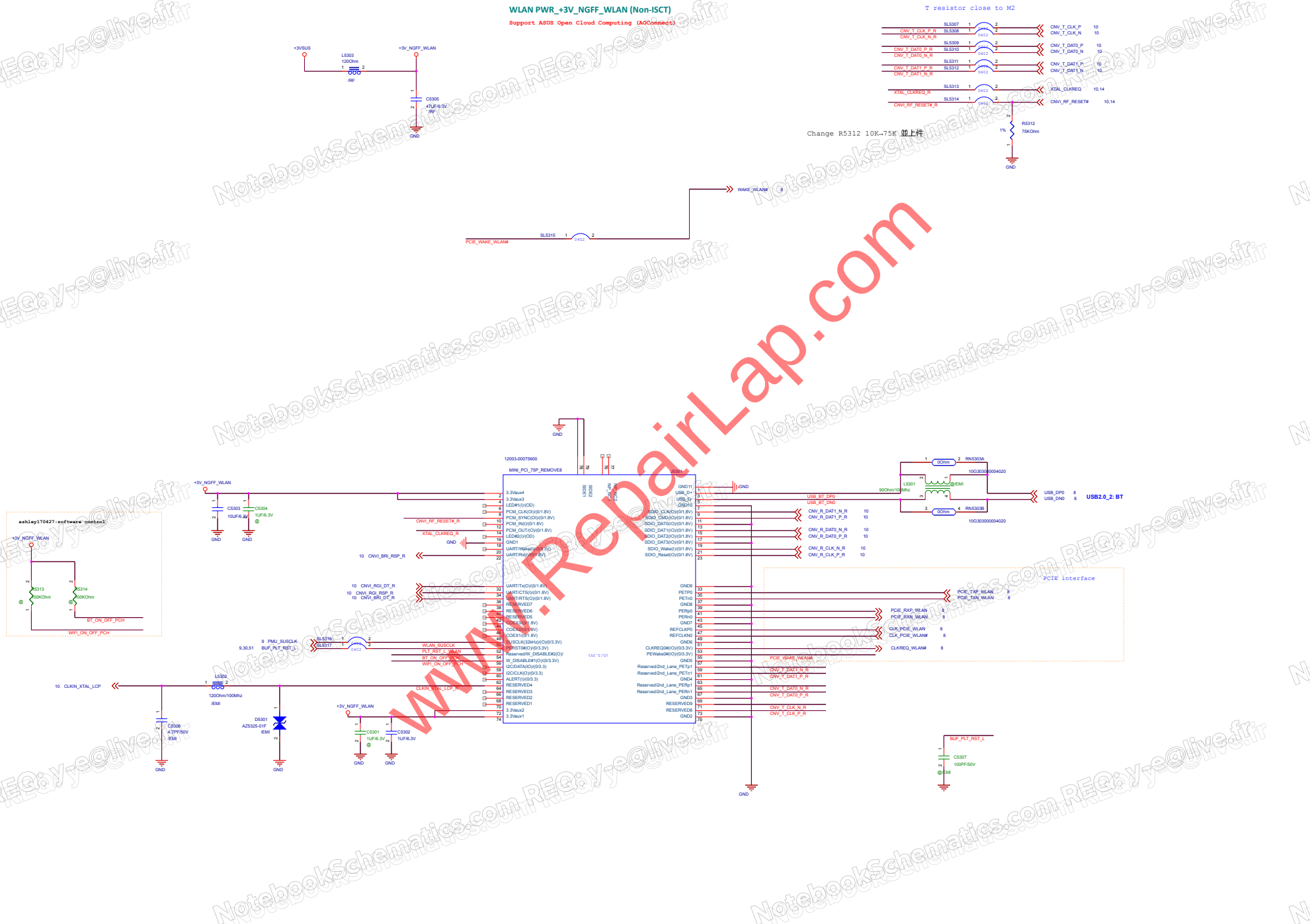

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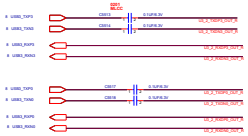
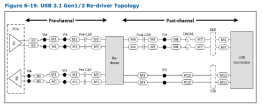
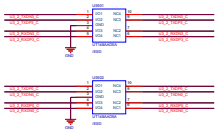


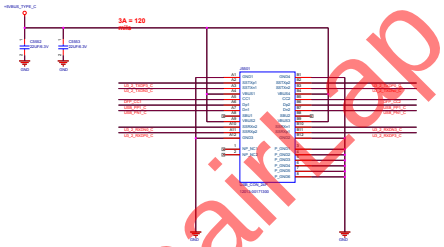


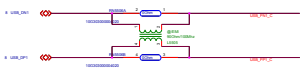
EM PASS 09022-00110000
CH PN
10Ω/1Ω/0000054039 RES A-S
OHM(0402)JUMP 2R4P/YAGEO

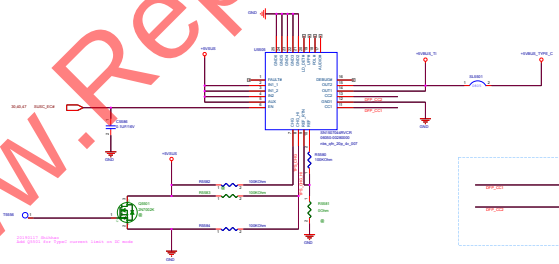
09022-00110000 PCB footprint option:
EMF mounted -> nbs_filter_8p_50fc
RN mounted -> Only RN circuit

USB3.0
ESD-Protection

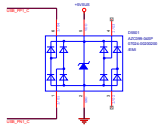








USB2.0
ESD-Protection



1st Source: P/N:07024-00200200 AMAZING/AZC099-04SP.R7G
2nd Source: P/N:07024-00710000 NXP/PUSB2X4D

CHG	CHG_HI	CC Capability Broadcast	Current Limit	Load Detect Threshold
0	0	STD	1.67 A	NA
0	1	STD	1.67 A	NA
1	0	1.5 A	1.67 A	NA
1	1	3.0 A	3.34 A	1.77 A

PCB Thickness & Layer	EMPass Model	ASUS P/N
0.8mm_4Layer	EU04AA1018AA	temp_T_001428
1.0mm_4Layer	EU04BB1018AA	temp_T_001448
1.0mm_6Layer 1.2mm_6Layer	EU06AA1018AA	temp_T_001430
1.0mm_8Layer 1.2mm_8Layer	EU08AA1018AA	temp_T_001429

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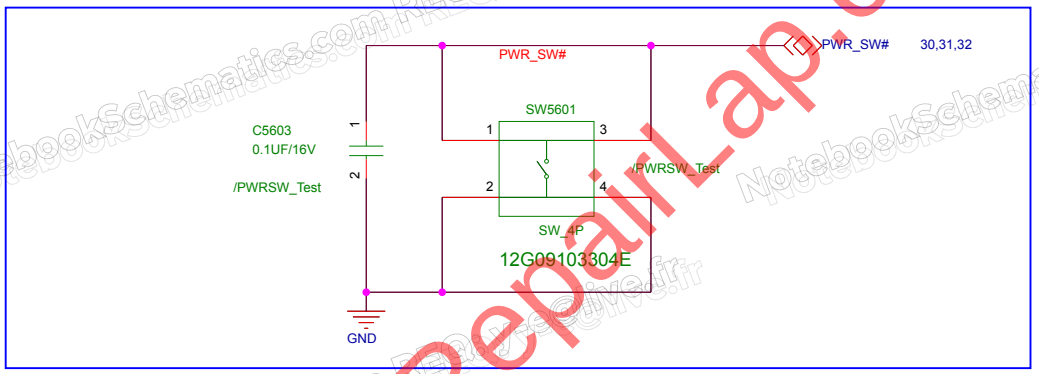

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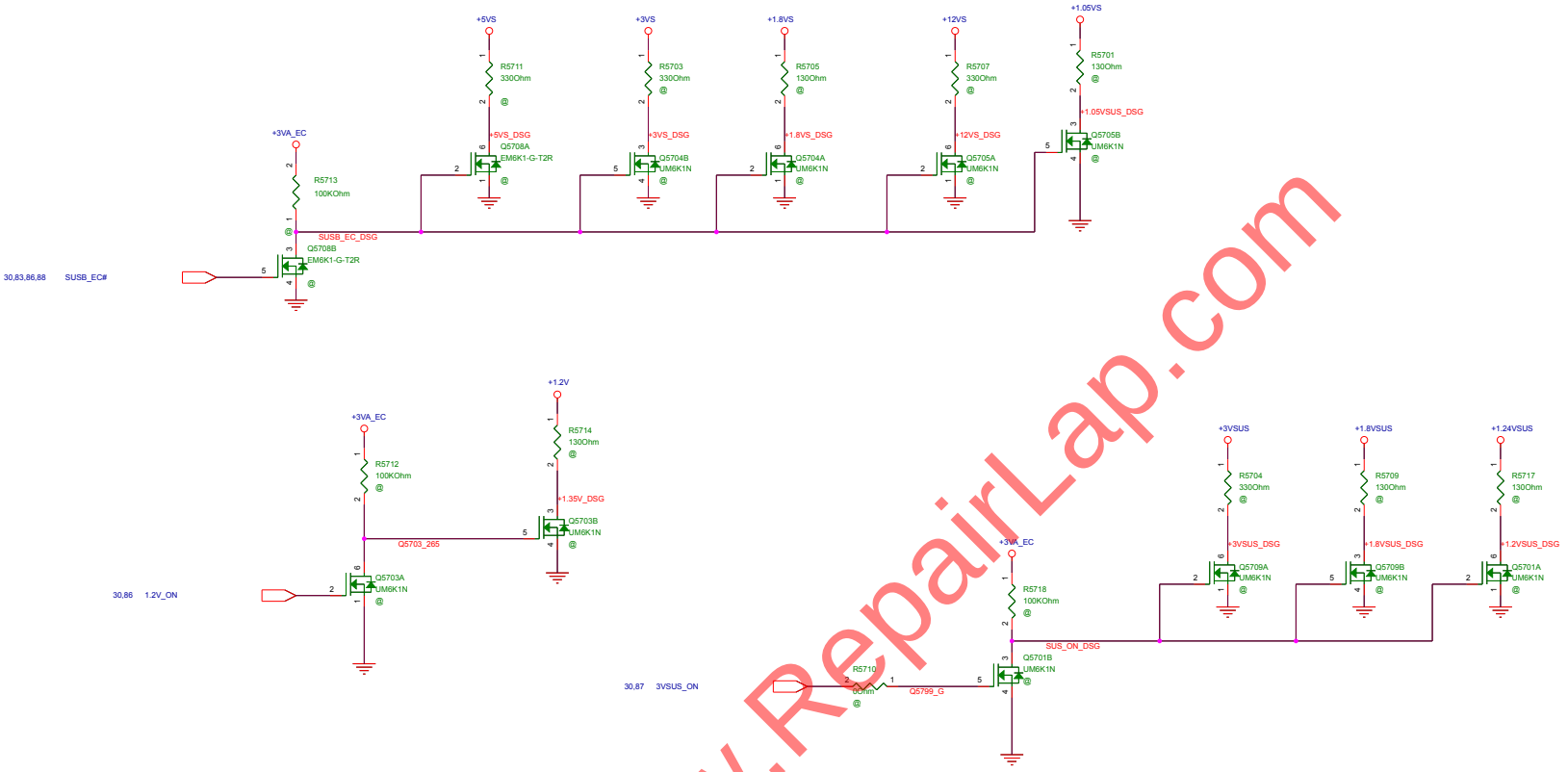
Main Board



<Variant Name>

ASUS	Project Name: X415MA	Rev: 0.9
Title : DSG_Discharge		
Size: B	Dept.: ASUSTeK COMPUTER INC.	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 57 of 99	

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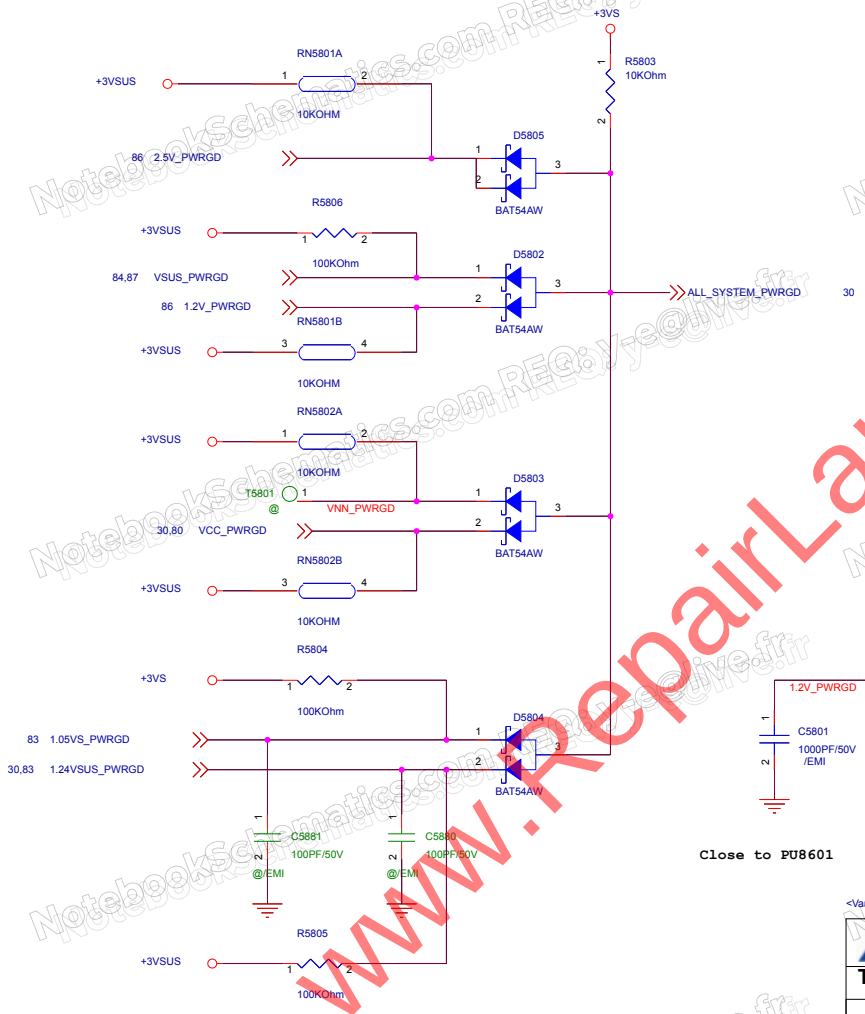

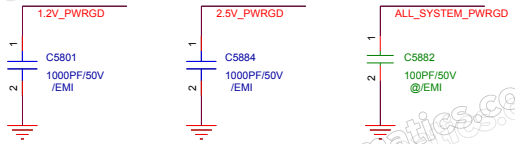

Close to PU8601

Close to PU8602

<Variant Name>

ASUS	Project Name: X415MA	Rev: 0.9
Title : PWR_Protect		
Size: Custom	Dept.: ASUSTeK COMPUTER INC.	Engineer: Mike
Date: Thursday, September 10, 2020	Sheet 58 of 99	

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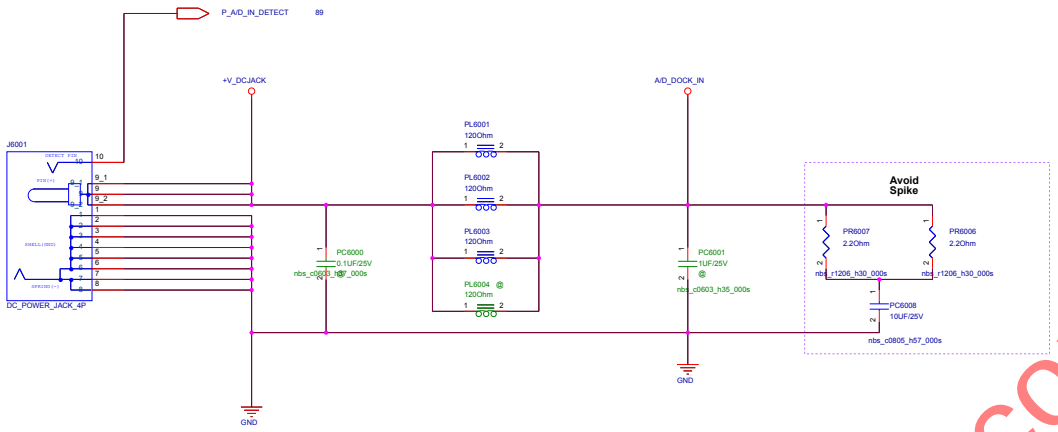
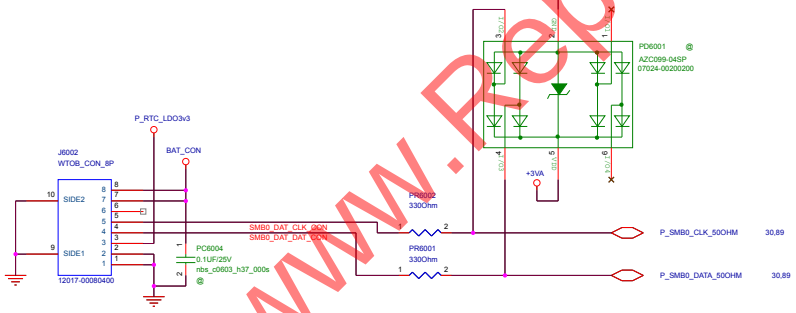
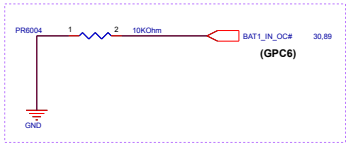


Battery Connector





Close to EC & CHECK EC Page Yes or No Pu Down

<Variant Name>

ASUS	Project Name: X415MA	Rev: R1.0
Title : DC & BAT Connector		
Size: Custom	Dept.: ASUS	Engineer: James
Date: Thursday, September 10, 2020	Sheet 60 of 102	

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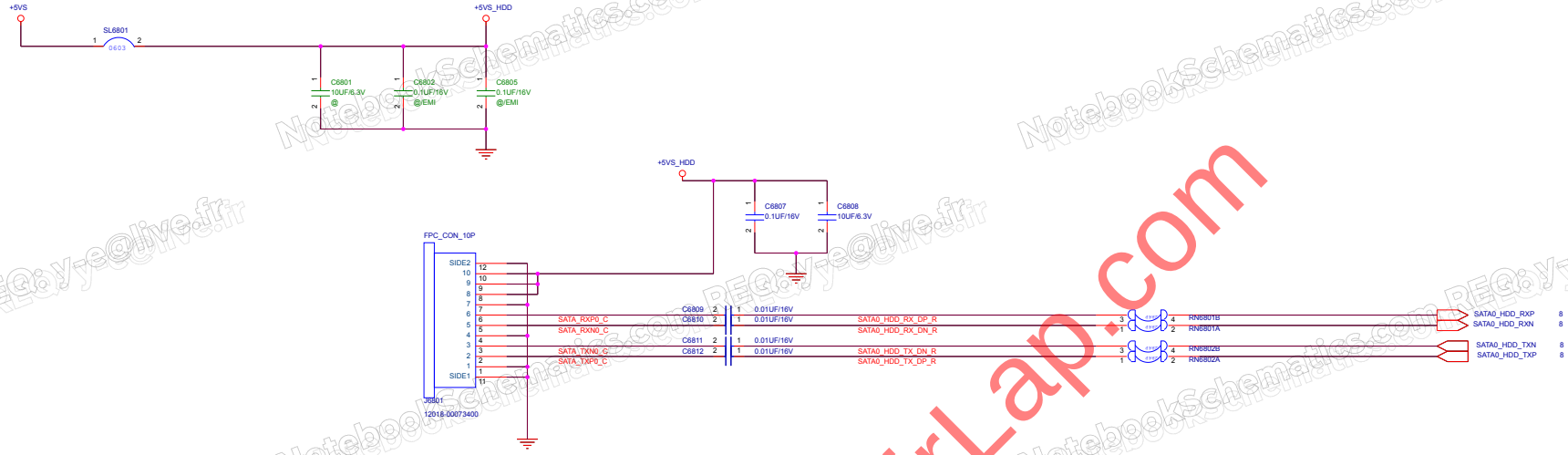
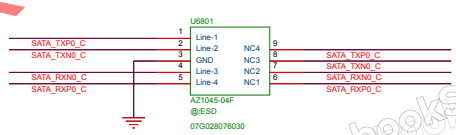


s530 FFC PIN DEFINE

FFC	PIN DEFINE	SATA CONN
	GND	P11
	GND	P10
1	5V	P9
2	5V	P8
3	5V	P7
	GND	P6
	GND	P5
	GND	P4
4	GND	S7
5	RXP(B+)	S6
6	RXN(B-)	S5
7	GND	S4
8	TXN(A-)	S3
9	TXP(A+)	S2
10	GND	S1



Reserve for EMI
170426 Shaun
Change to 07G028076030 ESD
170427 Shaun
4-5 6-7 swape
170428 Shaun

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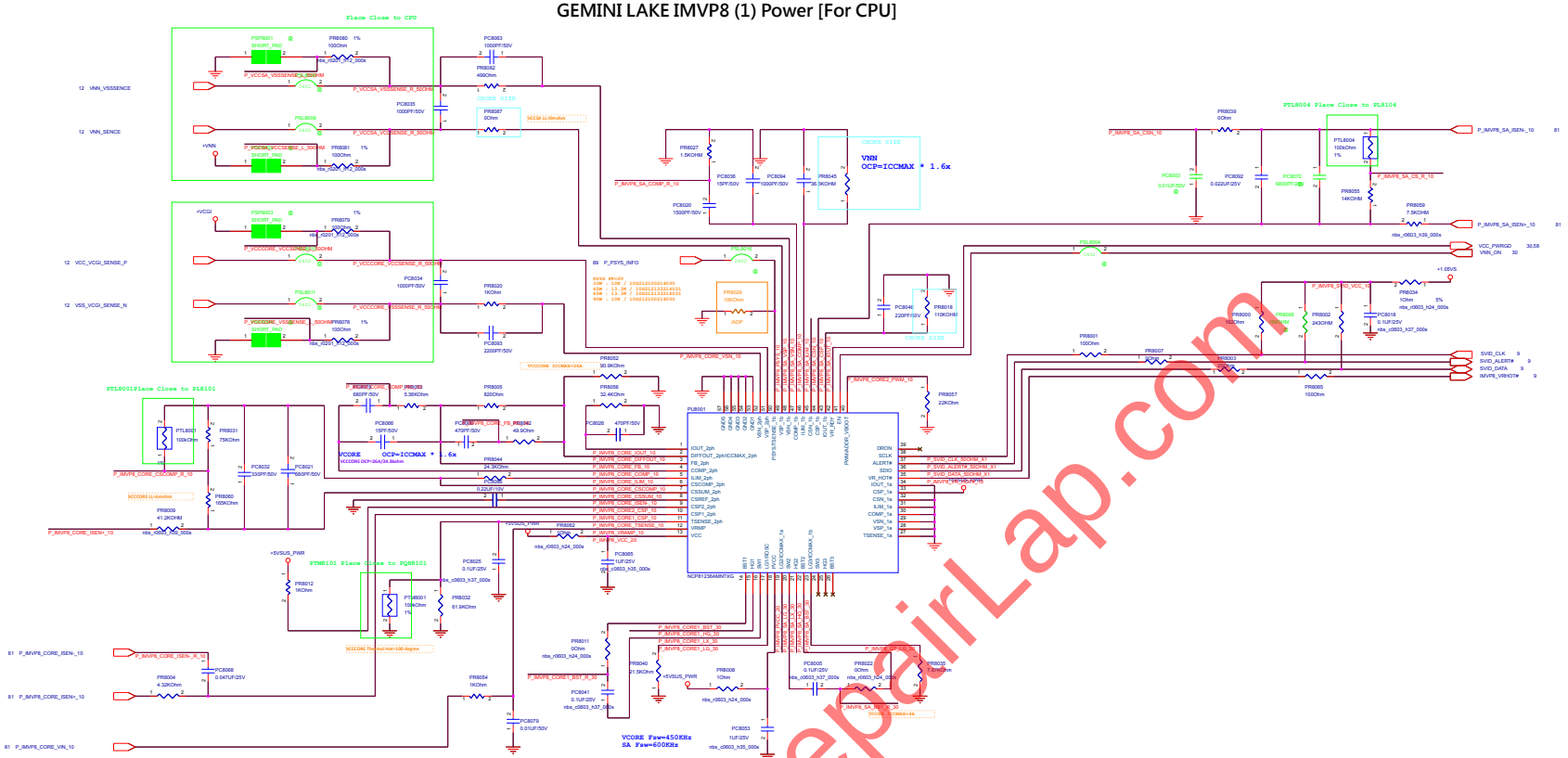
GEMINI LAKE IMVP8 (1) Power [For CPU]



NCP81236A Rpsys Setting

Adaptor	33W	45W	65W	90W
Psys_Pmax	100W	150W	150W	200W
CSG PROM Gain	2uA/W	1uA/W	1uA/W	1uA/W
Rpsys	10kohm	13.3kohm	13.3kohm	10kohm

Project Name	X415MA
Title :	PW_Gemini Lake IMVP8_(1)
Dept.:	NB Power Team
Engineer:	James
Date:	Thursday, September 10, 2020
Sheet	80 of 102

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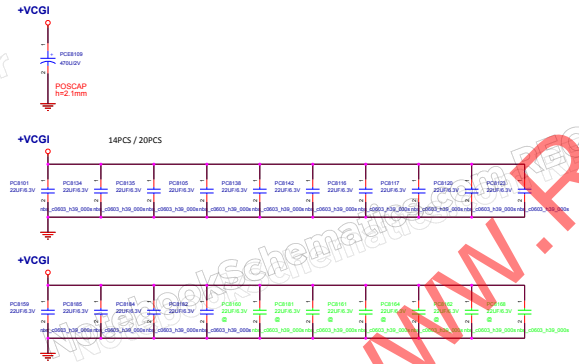
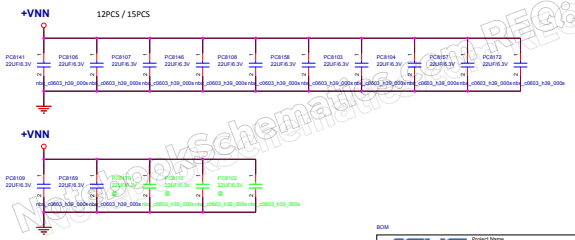
Gemini LAKE IMVP8 Power (2)[For CPU]





Project Name: X415MA	Rev: R1.0	
Title: PW_Gemini Lake IMVP8_(2)		
Size: Custom	Dept.: NB Power Team	Engineer: James
Date: Friday, September 11, 2020	Sheet: 81	of: 102

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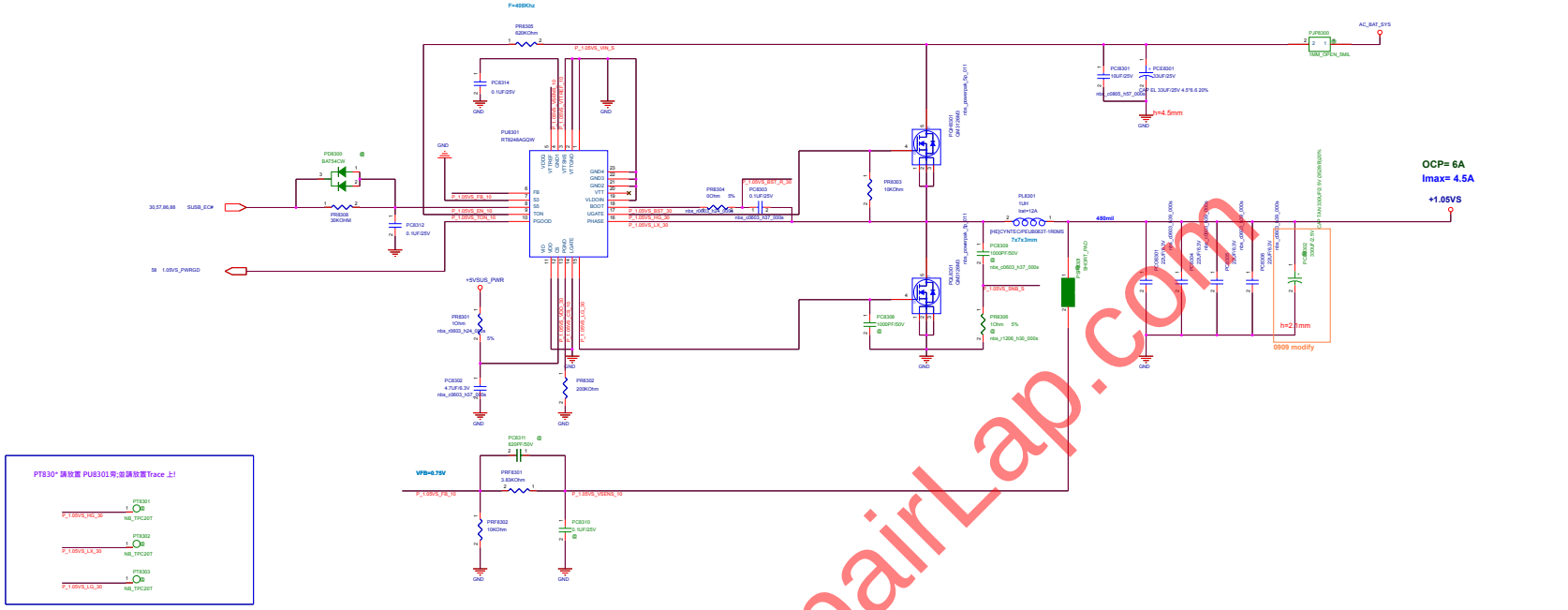
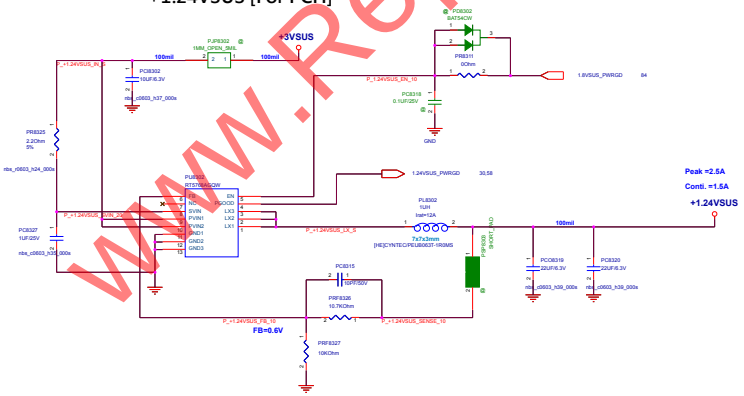
+1.05VS [For PCH]



+1.24VSUS [For PCH]



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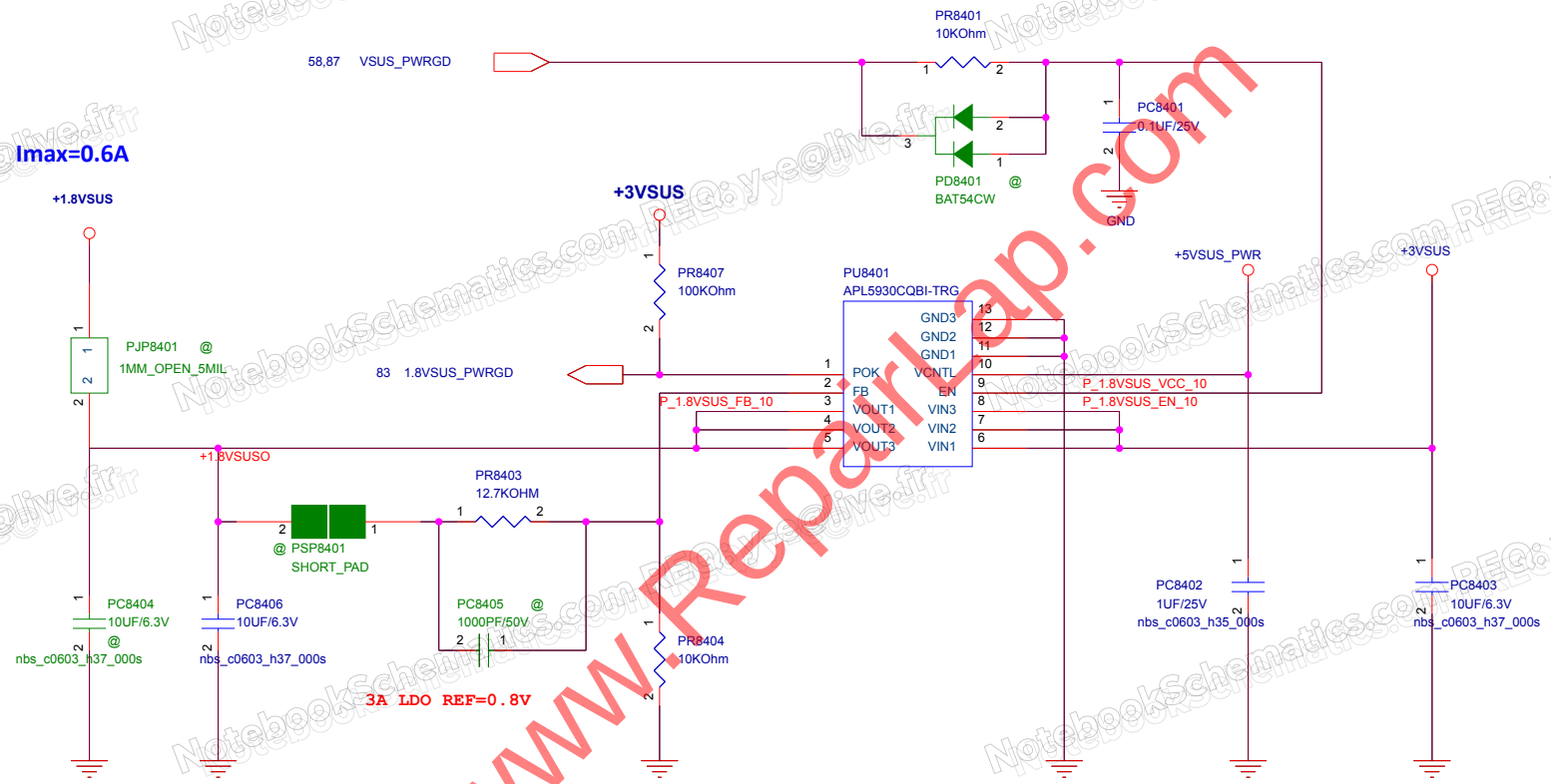
+1.8VSUS [For PCH]



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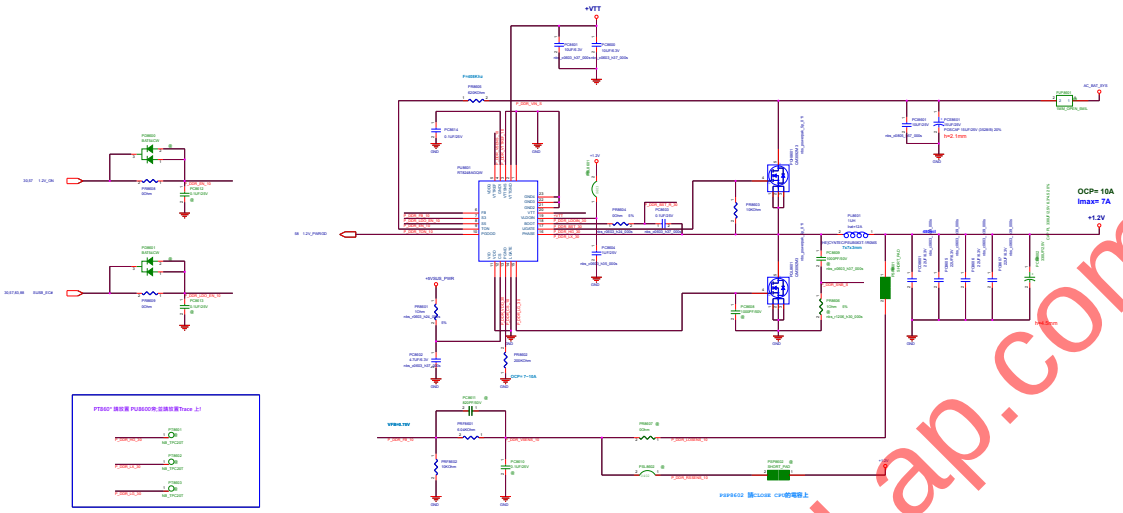
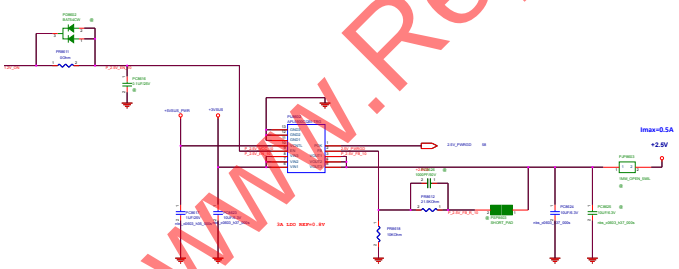
+1.2V / +VTT [For Memory]



+2.5V



ASUS	Project Name: X416MA	Rev: REV 1
Title :	PWR_VTT_VCCPIO_SUS	
Size:	Engineer: James	
Date: Tuesday, September 01, 2020	Sheet 55 of 102	

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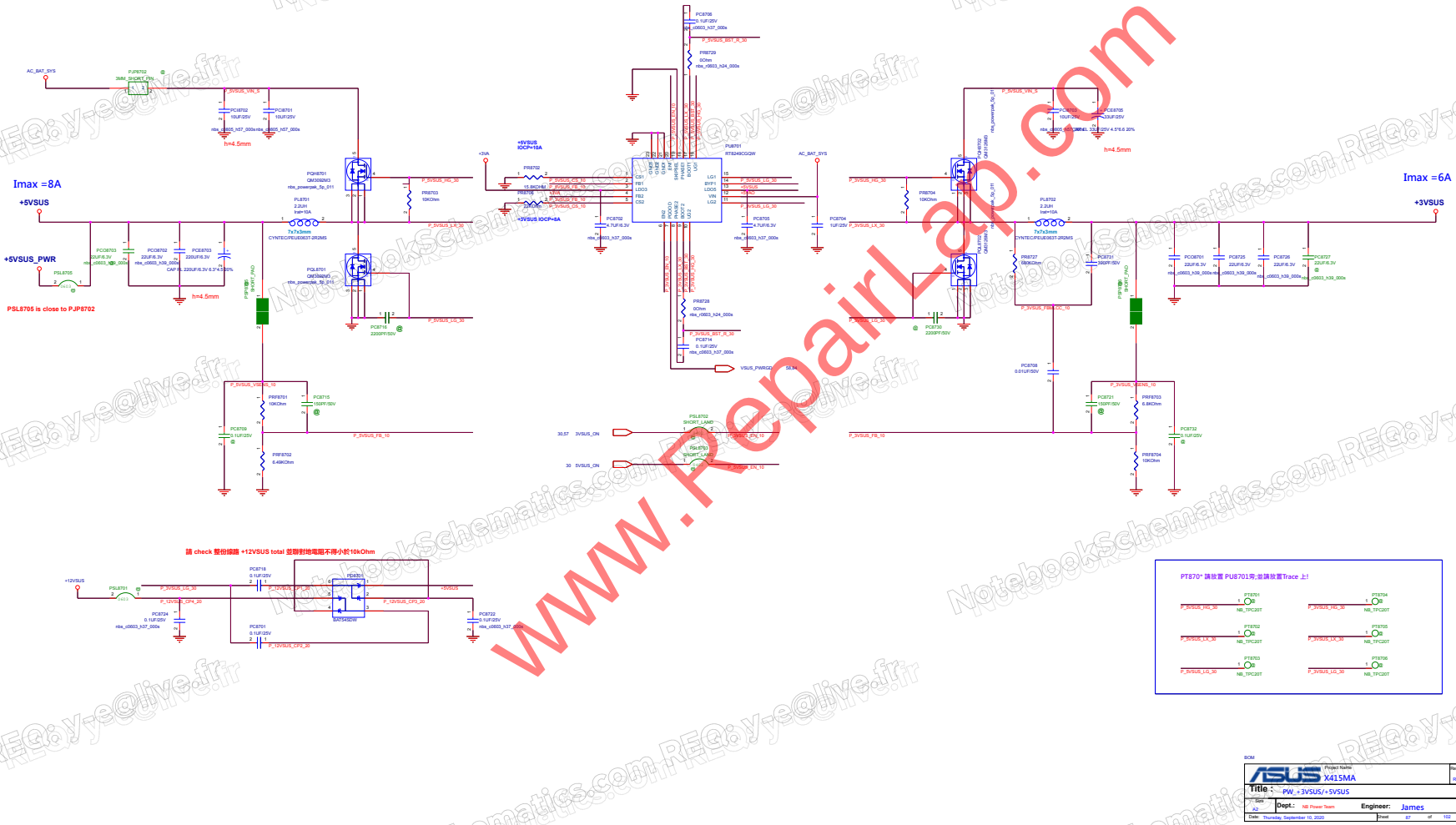
+3VSUS / +5VSUS [System Power]



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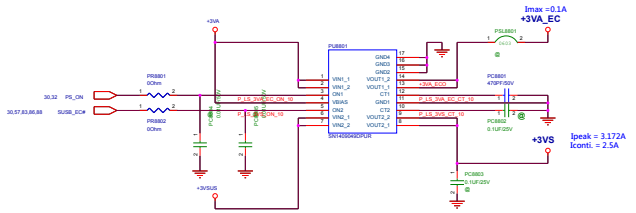
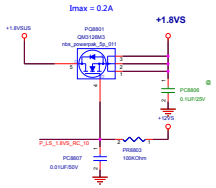
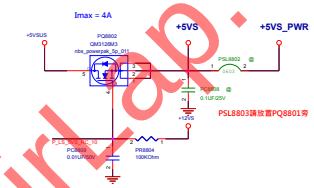
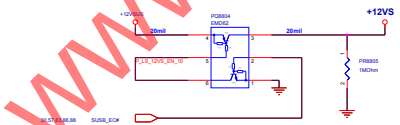
Project Name: X415MA			Rev: 80.1
Title: PW_+3VSUS/+5VSUS			
Size: A2	Dept.: NB Power Team	Engineer: James	
Date: Thursday, September 10, 2020	Sheet 87 of 102		

Load Switch









ASUS	Project Name: X415MA	Rev: 60.1
Title : PW_LOAD_SWITCH		
Size: A2	Dept.: NB Power Team	Engineer: Hector
Date: Thursday, September 10, 2020	Sheet 84 of 102	

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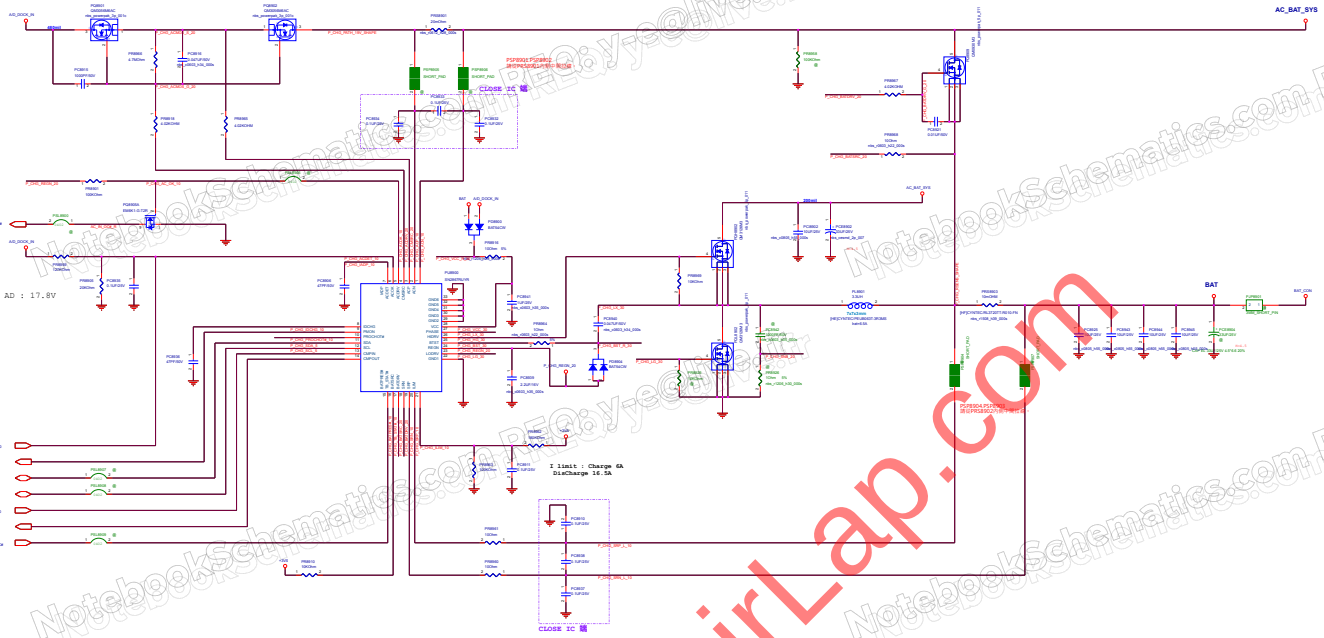
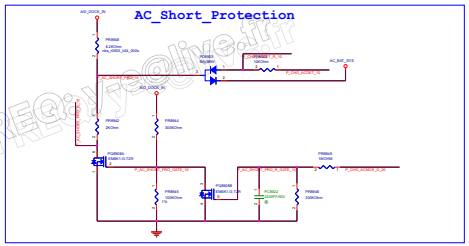
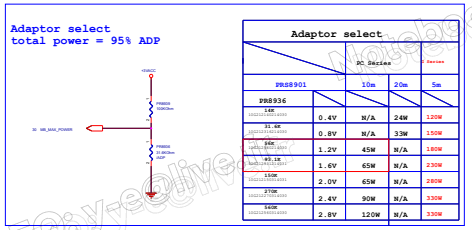


AC_Short_Protection



Adaptor select

total power = 95% ADP



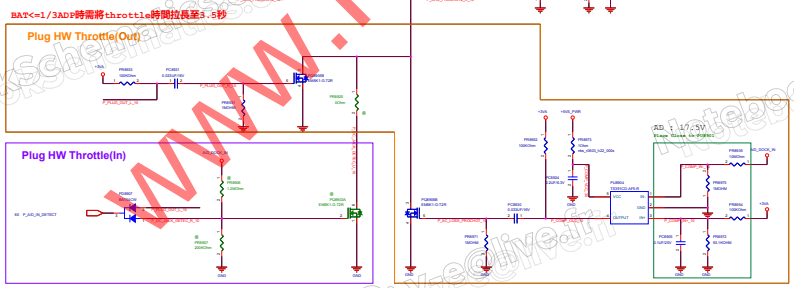
Adaptor select

PC_Series			
PR8901	10m	20m	5m
PR8936			
1.4K	0.4V	N/A	24W
21.0K	0.8V	N/A	33W
34K	1.2V	45W	N/A
49.1K	1.6V	65W	N/A
150K	2.0V	65W	N/A
270K	2.4V	90W	N/A
560K	2.8V	120W	N/A

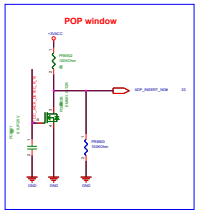
HW_Throttle

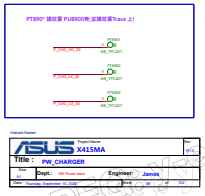
AC limit = 100% ADP
Bat limit = byte 7 x 1.7

BAT<=1/3ADP時需將throttle時間拉長至3.5秒



POP window





Title :	PW_CHARGER
Size :	X415MA
Dept :	NB Power team
Engineer :	Jamik

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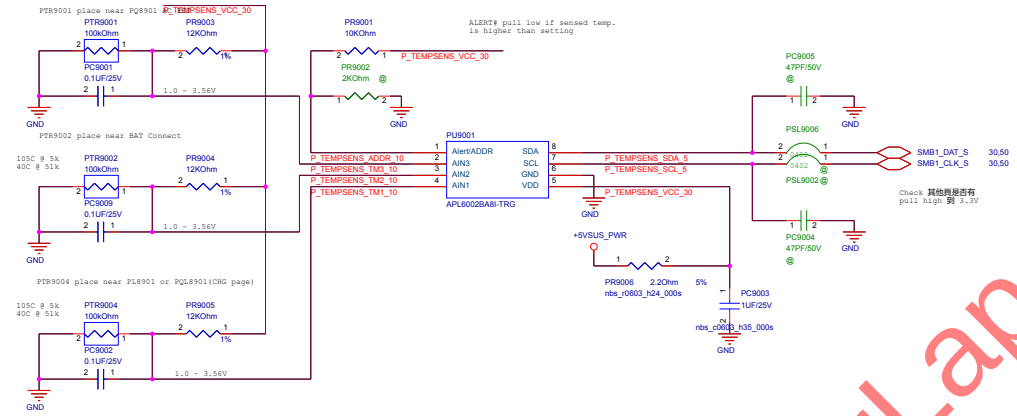
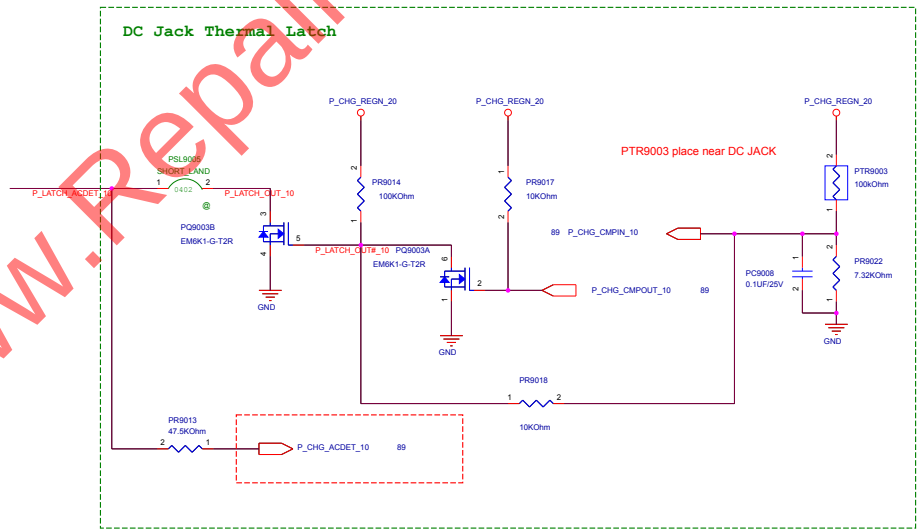
Address Selection Table

Address	0x7E	0x7C	0x7A	0x78	0x76	0x74	0x72	0x70
PR9001	10k	1.5k	2k	3.6k	3.9k	4.3k	5.1k	6k
PR9002	Open	8.2k	6.2k	6.8k	4.7k	3.6k	2.7k	2k

Register Address

Address	0x00	0x01	0x02	0x03	0x04	0x05	0x06
R/W	W	W	W	R	R	R	R
Function	Temp. alert threshold setting			Sensed temp. data			bit 4 = 0 bit 5 = 0 bit 6 = 0 When ALERT# assert





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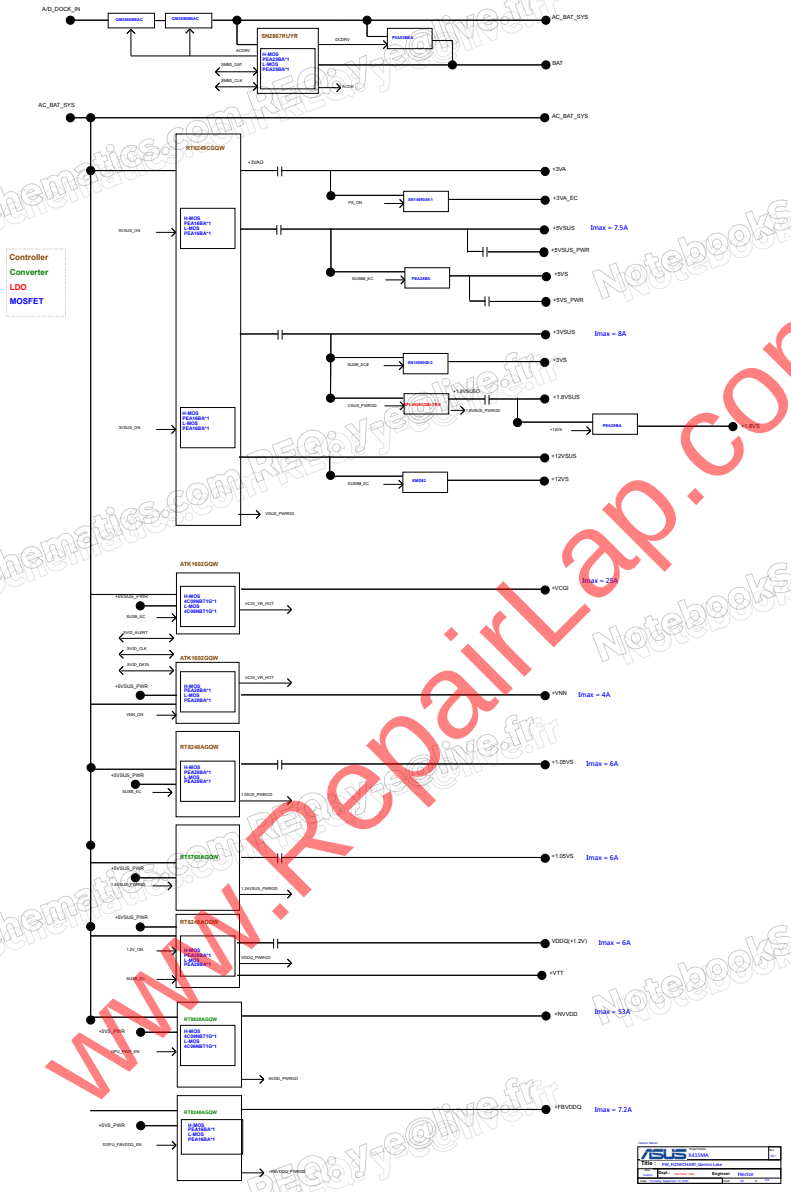

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